A Physical Introduction to Higher Mathematics

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Part I

Waves

1 From Superposition to Spectral Theory

Experimental observations.

- Small transverse disturbances on a taut string superpose: overlapping pulses add, and scaling an initial deflection scales its later motion. A finite string fixed at both ends has discrete resonant modes, while a pulse on an effectively infinite string requires a continuous range of frequencies.

1.1 Linear Structure and Superposition

Definition 1.1 (Vector spaces and linear maps)**.** Let k be a field. A *k-vector space* is an abelian group V with an action $k \times V \to V$ satisfying distributivity, associativity, and $1v = v$. A map $T : V \to W$ is *linear* if $T(av + bw) = aT(v) + bT(w)$.

Exercise 1.1 (String profiles and superposition)**.** Let V be the real-valued functions on $[0, L]$. Prove that the fixed-end profiles u satisfying $u(0) = u(L) = 0$ form a vector subspace. Express the opening string observation as closure under addition and scalar multiplication. Show that sampling a profile at finitely many points defines a linear map $V \to \mathbb{R}^n$.

Definition 1.2 (Subspaces, kernels, images, and quotients)**.** A *subspace* of V is a subset closed under linear combinations. For a linear map $T : V \to W$, define

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For a subspace $U \subseteq V$, the *quotient* V/U is the vector space of cosets $v + U$. The *direct sum* $V \oplus W$ is $V \times W$ with componentwise operations.

Exercise 1.2 (Kernels, images, and quotients)**.** Prove that $\ker T$ and $\operatorname{im} T$ are subspaces. Prove that a linear map $f : V \to X$ factors uniquely through V/U exactly when $U \subseteq \ker f$. Deduce $V/\ker T \cong \operatorname{im} T$.

Exercise 1.3 (Direct sums and isomorphism theorems)**.** Prove the product and coproduct universal properties of $V \oplus W$. For subspaces $A, B \subseteq V$, prove $A/(A \cap B) \cong (A + B)/B$. Deduce the remaining vector-space isomorphism theorems by applying this identity to nested subspaces.

Definition 1.3 (Duals and pairings)**.** The *dual* of V is $V^* = \operatorname{Hom}_k(V, k)$. A *pairing* of V and W is a bilinear map $V \times W \to k$. It is *nondegenerate* if each nonzero argument pairs nontrivially with some argument in the other space.

Definition 1.4 (Tensor products)**.** A *tensor product* of V and W is a vector space $V \otimes W$ with a bilinear map $(v, w) \mapsto v \otimes w$ such that every bilinear map $b : V \times W \to X$ factors uniquely through a linear map $\tilde{b} : V \otimes W \to X$. Tensor powers and multilinear maps are defined by iteration.

Definition 1.5 (Finite-dimensional duality, trace, and dimension)**.** The *evaluation* map is

$$\mathrm{ev}_V : V^* \otimes V \longrightarrow k, \qquad \varphi \otimes v \longmapsto \varphi(v).$$

The vector space V is *finite-dimensional* if there is a coevaluation map $\mathrm{coev}_V : k \to V \otimes V^*$ satisfying the snake identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{V^*} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^*}.$$

Let $s_{V,V^*} : V \otimes V^* \to V^* \otimes V$ exchange the tensor factors. For $T \in \mathrm{End}(V)$, define

$$\mathrm{tr}(T) = \mathrm{ev}_V \circ s_{V,V^*} \circ (T \otimes \mathrm{id}_{V^*}) \circ \mathrm{coev}_V \in k, \qquad \dim V = \mathrm{tr}(\mathrm{id}_V).$$

Exercise 1.4 (Duality, trace, and dimension)**.** Prove directly from the snake identities that the coevaluation is unique and that dualizable vector spaces are closed under direct sums, tensor products, and duals. For finite-dimensional V, W, prove $\mathrm{tr}(gf) = \mathrm{tr}(fg)$ for maps $f : V \to W$ and $g : W \to V$, and prove

$$\dim(V \oplus W) = \dim V + \dim W, \qquad \dim(V \otimes W) = \dim V \, \dim W.$$

Carry out these proofs using evaluation, coevaluation, symmetry, and their universal properties, without choosing coordinates.

Exercise 1.5 (Duality and multilinearity)**.** For finite-dimensional spaces, construct the natural maps

$$V^* \otimes W \longrightarrow \mathrm{Hom}(V, W), \qquad V^* \otimes W^* \longrightarrow (V \otimes W)^*.$$

Prove that they are isomorphisms, identify the evaluation pairing, and prove the tensor–hom adjunction $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W))$.

Definition 1.6 (Exterior powers, determinant, and characteristic polynomial)**.** The *tensor algebra* and *exterior algebra* of V are

$$T(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \qquad \Lambda V = T(V)/\langle v \otimes v : v \in V \rangle;$$

write $\Lambda^r V$ for the degree-r component. For finite-dimensional V, define its *rank* to be the unique integer $n \geq 0$ such that $\Lambda^n V \neq 0$ and $\Lambda^{n+1} V = 0$; the exercise below proves that this integer exists and that $\Lambda^n V$ is one-dimensional. For $T \in \mathrm{End}(V)$, the induced map $\Lambda^n T$ is multiplication by a scalar, the *determinant* $\det(T)$. The *characteristic polynomial* is

$$\chi_T(t) = \det(t \, \mathrm{id}_V - T).$$

Its root multiplicity at λ is the *algebraic multiplicity* of λ; the dimension of $\ker(T - \lambda \, \mathrm{id}_V)$ is its *geometric multiplicity.*

Exercise 1.6 (Top exterior powers and determinant identities)**.** Using only the universal property of alternating maps and the duality maps of V, prove that the rank n exists, that $\Lambda^n V$ is one-dimensional, and that $\Lambda^r V = 0$ for $r > n$. Prove

$$\det(ST) = \det(S) \det(T), \qquad \det(T \oplus R) = \det(T) \det(R).$$

Relate the scalar dimension to rank by $\dim V = (\mathrm{rank}\, V) 1_k$ and explain why the scalar dimension alone need not distinguish ranks in positive characteristic. For $V = \mathbb{R}^n$, identify $|\det T|$ with the volume-scaling factor and the sign with orientation. Deduce the change-of-variables rule for parallelepipeds and Cramer's rule.

1.2 Inner Products and Finite Spectral Theory

An inner product turns linear superposition into a geometry of orthogonal modes and makes spectral decomposition possible.

Definition 1.7 (Inner products and orthogonality)**.** An *inner product* on a real vector space is a positive-definite symmetric bilinear form. On a complex vector space it is a positive-definite sesquilinear form, conjugate-linear in the first variable. Write $\|v\| = \sqrt{\langle v, v\rangle}$. Vectors are *orthogonal* when their inner product is zero, and $U^\perp = \{v : \langle u, v\rangle = 0 \text{ for every } u \in U\}$.

Definition 1.8 (Projections and adjoints)**.** A *projection* is an endomorphism P with $P^2 = P$. An *orthogonal projection* is a projection with $\ker P = (\operatorname{im} P)^\perp$. The *adjoint* of $T : V \to W$ is a map $T^* : W \to V$ satisfying $\langle Tv, w\rangle = \langle v, T^*w\rangle$.

Exercise 1.7 (Orthogonal decomposition and best approximation)**.** Let V be finite-dimensional and $U \subseteq V$. Prove $V = U \oplus U^\perp$, construct the orthogonal projection P_U, and prove that $P_U v$ is the unique element of U minimizing $\|v - u\|$. Prove that P is an orthogonal projection if and only if $P = P^* = P^2$.

Definition 1.9 (Isometries and spectral classes)**.** A linear map is *orthogonal* over $\mathbb{R}$, or *unitary* over $\mathbb{C}$, if it preserves the inner product. An endomorphism A is *self-adjoint* if $A = A^*$, *normal* if $AA^* = A^*A$, and *positive* if $A = A^*$ and $\langle v, Av\rangle \geq 0$ for every v. A scalar λ is an *eigenvalue* if $\ker(A - \lambda \operatorname{id}) \neq 0$; a subspace U is *invariant* if $A(U) \subseteq U$. In finite dimensions, $\operatorname{spec}(A)$ denotes the set of eigenvalues of A in the scalar field.

Definition 1.10 (Complete commuting spectral projections)**.** A family (P_λ) of commuting orthogonal projections is *complete and mutually orthogonal* if

$$P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda, \qquad \sum_\lambda P_\lambda = \operatorname{id}.$$

A *spectral decomposition* of A is such a family satisfying

$$A = \sum_{\lambda \in \operatorname{spec}(A)} \lambda P_\lambda.$$

Exercise 1.8 (Finite-dimensional spectral theorem)**.** Prove that adjoints exist uniquely. Prove that a complex endomorphism is normal if and only if it admits a complete family of mutually orthogonal commuting spectral projections. Deduce the reality of self-adjoint spectra, the unit modulus of unitary spectra, and nonnegative spectra for positive operators. Prove that every positive operator A has a unique positive square root $A^{1/2}$ by applying the square-root function to its spectral projections. For a map $T : V \to W$ between finite-dimensional inner-product spaces, let (Q_s) be the complete spectral projections of $|T| = (T^*T)^{1/2}$. For each $s > 0$, construct an isometric isomorphism $U_s : Q_s V \to T(Q_s V)$ satisfying $TQ_s = sU_sQ_s$, and obtain the coordinate-free singular-value decomposition

$$T = \sum_{s>0} sU_sQ_s.$$

Exercise 1.9 (Commuting normal operators)**.** Prove that every spectral projection of a normal operator is a polynomial in that operator and its adjoint. For a finite commuting family of normal operators, construct a single complete family of mutually orthogonal commuting projections (P_α) and scalars $\lambda_j(\alpha)$ such that

$$A_j = \sum_\alpha \lambda_j(\alpha) P_\alpha$$

for every member A_j of the family. Interpret $P_\alpha V$ as the joint spectral subspaces without choosing coordinates.

1.3 Oscillators and Normal Modes

The finite spectral theorem turns coupled equations of motion into independent normal modes. A chain of masses provides a finite approximation to the same decomposition for a continuous string.

Definition 1.11 (Linear oscillator)**.** A *linear oscillator* on a finite-dimensional real inner-product space V is an equation

$$M\ddot{q} + Kq = 0,$$

where M is positive definite and K is self-adjoint and positive semidefinite. A *normal mode* is a nonzero $v \in V$ for which $q(t) = a(t)v$ solves the equation for some nonconstant a.

Exercise 1.10 (The harmonic oscillator)**.** Solve $m\ddot{q} + kq = 0$ for $m, k > 0$. Prove conservation of $E = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2$, determine the period, and deduce from the measured period of a small-amplitude mass–spring system the ratio k/m.

Exercise 1.11 (Critical damping and a repeated root)**.** For $m, b, \kappa > 0$, apply the exponential ansatz to $m\ddot{x} + b\dot{x} + \kappa x = 0$ and solve the equation in the underdamped, overdamped, and critically damped regimes. Obtain the second critical solution $te^{-bt/(2m)}$ directly from the repeated-root limit. Show that critical damping is exactly $b^2 = 4m\kappa$. Among $b^2 \geq 4m\kappa$, show that the critical value maximizes the magnitude of the slow decay rate, explaining the fastest asymptotic return without oscillatory eigenvalues.

Exercise 1.12 (Coupled oscillators and normal modes)**.** Set $x = M^{1/2}q$ and $L = M^{-1/2}KM^{-1/2}$. Use the spectral projections of L to derive the normal-mode decomposition. For a fixed-end chain of N equal masses joined by equal springs, compute the mode shapes and frequencies and relate them to the modes of a taut string.

Definition 1.12 (Vibrating string)**.** For length L, tension τ, and linear density ρ, a *vibrating string* is a function $u : [0, L] \times \mathbb{R} \to \mathbb{R}$ satisfying fixed-end conditions $u(0, t) = u(L, t) = 0$ and

$$\rho\, \partial_t^2 u = \tau\, \partial_x^2 u.$$

Exercise 1.13 (String modes and measured pitch)**.** Derive the vibrating-string equation from transverse force balance in the small-slope limit. Determine all separated normal modes and their frequencies. Prove that the fundamental frequency is $f_1 = (2L)^{-1}\sqrt{\tau/\rho}$ and compare the predicted ratios $f_n/f_1 = n$ with the harmonics of a stretched string.

1.4 Fourier Analysis and Continuum Modes

For a continuous string, finite sums of normal modes give way to orthogonal series; on an unbounded domain, the mode parameter becomes continuous. The required setting is furnished by measure and integration.

Definition 1.13 (Measure spaces and simple functions)**.** A *measure space* is a triple (X, Σ, μ) in which Σ is a σ-algebra and $\mu : \Sigma \to [0, \infty]$ is countably additive on disjoint families, with $\mu(\varnothing) = 0$. A function is *measurable* when inverse images of Borel sets are in Σ. A nonnegative *simple function* is a finite sum $\sum_j a_j 1_{E_j}$ with $a_j \geq 0$ and $E_j \in \Sigma$.

Definition 1.14 (Fourier function spaces)**.** For $1 \le p < \infty$, the space $L^p(X, \mu)$ consists of measurable functions modulo equality almost everywhere with

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p} < \infty;$$

L^∞ uses the essential supremum. For Lebesgue measure on $\mathbb{R}^n$ and normalized arc measure on $\mathbb{T}$, write $L^p(\mathbb{R}^n)$ and $L^p(\mathbb{T})$. The inner product on $L^2(\mathbb{T})$ is $\langle f, g\rangle = (2\pi)^{-1} \int_0^{2\pi} \overline{f(x)} g(x)\, dx$. The exercises below establish the analytic properties used in this section.

Exercise 1.14 (Integration and monotone convergence)**.** Define the integral of a nonnegative simple function and then of an arbitrary nonnegative measurable function by monotone approximation. Prove monotone convergence and Fatou's lemma.

Exercise 1.15 (Dominated convergence and product measures)**.** Construct the integral of an integrable complex function from its positive and negative real and imaginary parts. Prove dominated convergence. Construct product measure for σ-finite spaces and prove Tonelli's and Fubini's theorems.

Exercise 1.16 (L^p inequalities)**.** Prove Hölder's and Minkowski's inequalities for $1 \le p < \infty$.

Exercise 1.17 (L^p completeness and approximation)**.** Prove completeness of L^p for $1 \le p < \infty$. For Lebesgue measure, prove density first of simple functions with finite-measure support and then of $C_c(\mathbb{R}^n)$. Deduce that the inner-product norm makes L^2 complete.

Definition 1.15 (Orthogonal expansions)**.** Let H be an inner-product space. For an orthogonal family (e_n), the *orthogonal coefficients* of f are $\widehat{f}(n) = \langle e_n, f\rangle / \|e_n\|^2$. A *Fourier series* on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ is the formal expansion $f \sim \sum_{n\in\mathbb{Z}} \widehat{f}(n) e^{inx}$, where $\widehat{f}(n) = (2\pi)^{-1} \int_0^{2\pi} f(x) e^{-inx}\, dx$.

Exercise 1.18 (Bessel and Parseval criteria)**.** Prove Bessel's inequality for every finite orthogonal family. Prove that an orthonormal family (e_n) is complete precisely when

$$\|f\|^2 = \sum_n |\langle e_n, f\rangle|^2$$

for every f.

Exercise 1.19 (Completeness of Fourier modes)**.** Construct the Fejér kernels, prove that they form an approximate identity, and use convolution with them to prove completeness of $(e^{inx})_{n\in\mathbb{Z}}$ in $L^2(\mathbb{T})$. Deduce Parseval's identity and L^2 convergence of Fourier series.

Exercise 1.20 (Spectral evolution of a string)**.** For smooth fixed-end initial displacement and velocity satisfying the endpoint compatibility conditions, expand the data in sine modes and derive the classical solution of the wave equation. Prove uniqueness and conservation of wave energy from the spectral coefficients.

Exercise 1.21 (Spectral evolution of heat)**.** For smooth fixed-end initial temperature, use the same sine modes to derive the heat-equation solution $\partial_t u = \kappa \partial_x^2 u$. Prove uniqueness and monotone decay of L^2 energy from the spectral coefficients.

Definition 1.16 (Schwartz functions, Fourier transform, and convolution)**.** A *Schwartz function* on $\mathbb{R}^n$ is a smooth function f such that $\sup_x |x^\alpha \partial^\beta f(x)| < \infty$ for every pair of multi-indices α, β. The

Schwartz space is denoted $\mathcal{S}(\mathbb{R}^n)$. For $f, g \in C_c(\mathbb{R}^n)$, define the *Fourier transform* and *convolution* by

$$\mathcal{F}f(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix\cdot\xi}\, dx, \qquad (f * g)(x) = \int_{\mathbb{R}^n} f(y) g(x-y)\, dy.$$

The estimates $\|\mathcal{F}f\|_\infty \le (2\pi)^{-n/2}\|f\|_1$ and $\|f * g\|_1 \le \|f\|_1\|g\|_1$ extend these operations to the L^1 completion.

Exercise 1.22 (Fourier inversion and convolution)**.** For Schwartz functions, prove Fourier inversion and $\mathcal{F}(f * g) = (2\pi)^{n/2}(\mathcal{F}f)(\mathcal{F}g)$.

Exercise 1.23 (Plancherel theorem)**.** Prove on $\mathcal{S}(\mathbb{R}^n)$ that $\mathcal{F}$ preserves the L^2 inner product. Prove that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ and extend $\mathcal{F}$ uniquely to a unitary operator and deduce Plancherel's identity.

Exercise 1.24 (Wave and heat propagators)**.** Use the Fourier transform to solve the Cauchy problems for

$$\partial_t^2 u = c^2 \Delta u, \qquad \partial_t u = \kappa \Delta u$$

on $\mathbb{R}^n$. Identify their spectral-mode evolution, derive the Gaussian heat kernel and its convolution law, and prove finite propagation speed for the one-dimensional wave equation.

1.5 Hilbert Spaces and Spectral Measures

Fourier series and transforms are concrete spectral decompositions. Hilbert space theory isolates the structure they share and replaces sums over modes by integration against projection-valued measures.

Definition 1.17 (Hilbert spaces and bounded operators)**.** In this subsection and the next, Hilbert spaces are complex. A *Hilbert space* is a complete inner-product space. A linear map $T : H \to K$ is *bounded* if $\|T\| = \sup_{\|x\|\le 1} \|Tx\| < \infty$. Write $B(H, K)$ for the space of bounded maps and $B(H) = B(H, H)$. The adjoint of T is the unique bounded map satisfying $\langle Tx, y\rangle = \langle x, T^*y\rangle$. Projections, self-adjoint operators, and unitary operators satisfy respectively $P = P^* = P^2$, $A = A^*$, and $U^*U = UU^* = \mathrm{id}$. For $T \in B(H)$, the *resolvent set* consists of $\lambda \in \mathbb{C}$ for which $T - \lambda\,\mathrm{id}$ has a bounded inverse; its complement is the *spectrum* $\mathrm{spec}(T)$.

Exercise 1.25 (Hilbert-space foundations)**.** Prove the projection theorem for closed subspaces and the Riesz representation theorem. Deduce $H = M \oplus M^\perp$ for closed M and prove existence, uniqueness, and $\|T^*\| = \|T\|$ for bounded adjoints.

Definition 1.18 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X assigns an orthogonal projection $E(B)$ to each measurable set, with $E(X) = \mathrm{id}$ and countable additivity in the strong operator topology. For bounded measurable f, the operator $\int_X f\, dE$ is defined by uniform approximation from simple functions.

Exercise 1.26 (Bounded spectral theorem)**.** Prove that every bounded self-adjoint operator A has a unique projection-valued measure on $\mathrm{spec}(A) \subset \mathbb{R}$ such that $A = \int \lambda\, dE(\lambda)$. Extend the statement to bounded normal operators, derive continuous functional calculus, and characterize unitary and projection spectra.

1.6 Unbounded Generators and Physical Spectra

The differential operators governing waves and translations are generally unbounded, so their domains become part of the spectral problem.

Definition 1.19 (Unbounded operators and domains)**.** A *densely defined operator* T has a dense linear domain $\mathrm{Dom}(T) \subset H$. It is *closed* when its graph is closed in $H \oplus H$; its graph norm is $\|x\|_T^2 = \|x\|^2 + \|Tx\|^2$. The domain of T^* consists of those y for which $x \mapsto \langle Tx, y\rangle$ is bounded in the Hilbert norm on $\mathrm{Dom}(T)$; for such y, the Riesz theorem gives the unique T^*y satisfying $\langle Tx, y\rangle = \langle x, T^*y\rangle$. The operator is *symmetric* when $\langle Tx, y\rangle = \langle x, Ty\rangle$ on its domain and *self-adjoint* when $T = T^*$, including equality of domains.

Definition 1.20 (Weak derivatives and first Sobolev spaces)**.** A function $g \in L^1_{\mathrm{loc}}(0, L)$ is the *weak derivative* of $f \in L^1_{\mathrm{loc}}(0, L)$ when

$$\int_0^L f\phi'\,dx = -\int_0^L g\phi\,dx \qquad (\phi \in C_c^\infty(0, L)).$$

The space $H^1(0, L)$ consists of the $f \in L^2(0, L)$ having weak derivative $f' \in L^2(0, L)$, with norm $\|f\|_{H^1}^2 = \|f\|_2^2 + \|f'\|_2^2$.

Exercise 1.27 (Completeness and representatives of H^1)**.** Prove that $H^1(0, L)$ is complete. Prove that every class has a unique absolutely continuous representative and that its classical derivative agrees almost everywhere with its weak derivative.

Exercise 1.28 (Traces and H_0^1)**.** Prove that endpoint evaluation on the absolutely continuous representative is continuous, establish integration by parts, and identify $H_0^1(0, L)$ as the closure of $C_c^\infty(0, L)$ and as the subspace with zero endpoint values.

Exercise 1.29 (Self-adjoint momentum and boundary holonomy)**.** On $L^2([0, L])$, let $P_\theta = -i\hbar\, d/dx$ on the absolutely continuous functions with derivative in L^2 and boundary condition $\psi(L) = e^{i\theta}\psi(0)$. Use integration by parts and the adjoint domain to prove that P_θ is self-adjoint. Determine its mutually orthogonal spectral projections $(E_n)_{n\in\mathbb{Z}}$, prove that they sum strongly to the identity, and show that their spectral values are

$$p_n = \frac{\hbar}{L}(2\pi n + \theta), \qquad n \in \mathbb{Z}.$$

Interpret θ as a boundary holonomy, later realized by magnetic flux through a ring, and explain why changing the domain changes the measured spectrum.

Exercise 1.30 (Cayley transform and spectral measure)**.** Use the Cayley transform and the bounded spectral theorem to prove that every self-adjoint H has a unique projection-valued measure with

$$H = \int_{\mathbb{R}} \lambda\, dE_H(\lambda), \qquad \mathrm{Dom}(H) = \left\{\psi : \int_{\mathbb{R}} \lambda^2\, d\langle \psi, E_H(\lambda)\psi\rangle < \infty\right\}.$$

Exercise 1.31 (Measurable functional calculus)**.** For a self-adjoint H with spectral measure E_H, define $f(H)$ first for bounded measurable f and then on the domain

$$\left\{\psi : \int_{\mathbb{R}} |f(\lambda)|^2\, d\langle \psi, E_H(\lambda)\psi\rangle < \infty\right\}.$$

Prove the algebraic, adjoint, and composition rules wherever the natural domains make them meaningful.

Exercise 1.32 (Stone's theorem)**.** Prove that $U_t = e^{-itH/\hbar}$ is a strongly continuous unitary group for every self-adjoint H. Conversely, recover a unique self-adjoint generator H from every strongly continuous unitary group and identify its domain by the strong derivative at $t = 0$.

Exercise 1.33 (Fourier spectra and free quantum evolution)**.** Use the Fourier transform to realize $-i\hbar\partial_j$ and $-\Delta$ as multiplication by $\hbar\xi_j$ and $|\xi|^2$ on their natural domains. Deduce self-adjointness, identify their spectral projections, and derive the free Schrödinger evolution for $H = -(\hbar^2/2m)\Delta$. Prove preservation of the domain, norm, and expected energy, and use this concrete multiplication-operator model to verify the preceding spectral and Stone theorems, including their domain statements. Identify the continuous energy spectrum and relate it to the continuum of outgoing-electron energies above an ionization threshold.

Exercise 1.34 (Spectral lines from unitary evolution)**.** If $H|n\rangle = E_n|n\rangle$, use $U_t = e^{-itH/\hbar}$ to show that $\langle A(t)\rangle$ contains frequencies $(E_m - E_n)/\hbar$. Explain why spectroscopic peaks measure energy differences and why line strengths depend on matrix elements between the corresponding spectral subspaces. Convert the Balmer wavelengths 656.3, 486.1, 434.0, and 410.2 nm to energy differences using $E = hc/\lambda$. Identify the levels E_n, rather than their differences, with atoms of the spectral measure of H, and identify the transition frequencies with spectral values of the Liouvillian $A \mapsto [H, A]/\hbar$.

Part II

Symmetry

2 From Symmetry to Harmonic Analysis

Experimental observations.

- Infrared spectra of gas-phase NH_3 contain four fundamental vibrational bands, while in NH_2D each band arising from a doubly degenerate NH_3 vibration splits into distinct bands.
- Monochromatic X-rays scattered by a crystal form sharp peaks only at selected directions determined by wavelength and lattice spacing.

2.1 Finite Groups and Representations

Definition 2.1 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Exercise 2.1 (The group isomorphism theorems)**.** Prove that kernels are normal and that a normal subgroup is the kernel of the quotient homomorphism. Prove the three isomorphism theorems for groups and the universal property of G/N.

Definition 2.2 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Exercise 2.2 (Orbit–stabilizer and class equation)**.** Construct a natural bijection $G/G_x \to Gx$. For finite G, deduce the orbit–stabilizer formula and the class equation from the conjugation action. Use these results to prove that every group of prime order is cyclic.

Definition 2.3 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Exercise 2.3 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 2.4 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 2.4 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h\rangle_G = \frac{1}{|G|} \sum_{g\in G} \overline{f(g)}h(g).$$

Exercise 2.5 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that $\langle \chi_U, \chi_V\rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Exercise 2.6 (Symmetry projections)**.** For an irreducible character χ of a finite group, prove that

$$P_\chi = \frac{\dim \chi}{|G|} \sum_{g\in G} \overline{\chi(g)}\rho(g)$$

is the projection onto the χ-isotypic subspace. Apply these projections to the displacement space of a molecule whose equilibrium configuration is preserved by G.

Exercise 2.7 (Molecular symmetry)**.** For the equilibrium geometry of NH_3, determine the C_{3v}-action on nuclear displacements, remove translations and rotations, and prove

$$V_{\mathrm{vib}} \cong 2A_1 \oplus 2E.$$

Deduce four fundamental frequencies, two of them doubly degenerate. For NH_2D, restrict to C_s and prove $E|_{C_s} \cong A' \oplus A''$. Deduce that symmetry no longer enforces either degeneracy and compare with the observed band splitting.

2.2 Lie Groups, Rotations, and Spin

Finite symmetry groups explain discrete degeneracies. Smooth symmetry groups add infinitesimal generators, continuous quantum numbers, and spin.

Definition 2.5 (Topological spaces)**.** A *topology* on X is a family of subsets containing $\varnothing, X$ and closed under arbitrary unions and finite intersections. A map is *continuous* when inverse images of open sets are open. A space is *Hausdorff* when distinct points have disjoint neighborhoods, *second countable* when its topology is generated by a countable family of open sets, and *compact* when every open cover has a finite subcover.

Definition 2.6 (Topological and smooth manifolds)**.** A *topological manifold* of dimension n is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth.

Definition 2.7 (Tangent and cotangent spaces)**.** The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying $v(fg) = f(p)v(g) + g(p)v(f)$. The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of $F : M \to N$ is $dF_p(v)(f) = v(f \circ F)$.

Exercise 2.8 (Intrinsic tangent vectors)**.** Prove that derivations at p, equivalence classes of smooth curves through p, and the usual coordinate tangent vectors are naturally isomorphic. Prove the chain rule $d(G \circ F)_p = dG_{F(p)} \circ dF_p$ and compute tangent spaces of level sets $F^{-1}(q)$ at points where dF is surjective.

Definition 2.8 (Lie groups and Lie algebras)**.** A *Lie group* is a smooth manifold G whose multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 2.9 (One-parameter subgroups and the exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 2.9 (Infinitesimal representations)**.** For a finite-dimensional smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \left.\frac{d}{dt}\right|_0 \rho(\exp tX).$$

Exercise 2.10 (Differentiating symmetry)**.** Prove that $d\rho$ preserves brackets and that $\rho(\exp X) = \exp(d\rho(X))$. Prove that a connected Lie-group representation is determined by its infinitesimal representation whenever the latter integrates uniquely.

Definition 2.10 (Rotations and spin)**.** Define

$$SO(3) = \{R \in \mathrm{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},$$
$$SU(2) = \{U \in \mathrm{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}.$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 2.11 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations and the 4π restoration observed by neutron interferometry.

Exercise 2.12 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \dots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \dots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$. For $j = \frac{1}{2}$, relate the two J_3 outcomes to the two Stern–Gerlach beams.

Exercise 2.13 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 2.11 (Quantum symmetries)**.** Let a compact group G act continuously and unitarily by $U : G \to U(H)$ on a Hilbert space with self-adjoint Hamiltonian $\mathcal{H}$. It is a *symmetry group* when $U_g \operatorname{Dom}(\mathcal{H}) = \operatorname{Dom}(\mathcal{H})$ and $\mathcal{H}U_g\psi = U_g\mathcal{H}\psi$ for every $g \in G$. A *quantum number* is an eigenvalue of a self-adjoint operator whose spectral projections commute with those of $\mathcal{H}$. A bounded operator multiplet A has *symmetry type* W when its components define an intertwiner from W to $B(H)$ under conjugation by G.

Exercise 2.14 (Symmetry, degeneracy, and selection rules)**.** Prove that every finite-dimensional eigenspace of $\mathcal{H}$ is symmetry-invariant and decomposes into irreducible G-representations. Deduce symmetry-forced degeneracies. Prove that a transition matrix element $\langle f, Ai \rangle$ can be nonzero only if the trivial representation occurs in $V_f^* \otimes W \otimes V_i$, and derive the angular-momentum rules for an electric-dipole operator.

Exercise 2.15 (Atomic spectra)**.** For the Coulomb Hamiltonian, use rotational symmetry and separation into irreducible $SO(3)$-types to obtain the quantum numbers ℓ, m and their degeneracies. With the radial spectrum $E_n = -m_e e^4/(2(4\pi\varepsilon_0)^2\hbar^2 n^2)$ as derived input, compute the Balmer wavelengths and compare them with the observed hydrogen lines. Apply the electric-dipole rule to identify forbidden transitions.

2.3 Haar Measure, Fourier Analysis, and Reciprocal Space

Invariant integration turns representations into a generalized Fourier analysis. For translation symmetry, its dual variables are the reciprocal coordinates measured by diffraction.

Definition 2.12 (Haar measure and convolution)**.** Let G be a locally compact Hausdorff group. A *left Haar measure* is a nonzero regular Borel measure μ satisfying $\mu(gE) = \mu(E)$. Haar's theorem gives such a measure, unique up to a positive scalar; on a compact group normalize it by $\mu(G) = 1$. For $f, g \in C_c(G)$, define

$$(f * g)(x) = \int_G f(y) g(y^{-1}x)\, d\mu(y).$$

The *left regular representation* on $L^2(G)$ is $(\lambda_g f)(x) = f(g^{-1}x)$.

Exercise 2.16 (Haar measure and the regular representation)**.** Identify Haar measure on $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$. Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the L^1 convolution inequality, and $\|f * g\|_2 \le \|f\|_1 \|g\|_2$. Prove that $g \mapsto \lambda_g$ is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over G to obtain an invariant one.

Definition 2.13 (Pontryagin duality)**.** For a locally compact abelian group G, its *Pontryagin dual* $\widehat{G}$ is the group of continuous characters $\chi : G \to U(1)$, with pointwise multiplication and the compact-open topology. For $f \in L^1(G)$, define

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, d\mu(x).$$

With compatible Haar measures on G and $\widehat{G}$, Fourier inversion and Plancherel extend the earlier theorems. Pontryagin duality identifies G naturally with $\widehat{\widehat{G}}$.

Exercise 2.17 (Abelian Fourier duality)**.** Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \qquad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \qquad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. Show that the group Fourier transform recovers Fourier transforms on $\mathbb{R}^n$ and Fourier series on $\mathbb{T}^n$. For a closed subgroup $H \leq G$, define its annihilator $H^\perp \subseteq \widehat{G}$ and prove $\widehat{G/H} \cong H^\perp$.

Exercise 2.18 (Interference and Fourier optics)**.** For coherent scalar waves with complex amplitudes a_1, a_2, derive $|a_1 + a_2|^2$ and the fringe spacing in Young's double-slit geometry. In the Fraunhofer regime, derive that the far-field amplitude is the Fourier transform of the aperture. Compute the single-slit diffraction intensity, locate its zeros, and compare the predicted angular spacing with measured optical diffraction.

Definition 2.14 (Lattices and reciprocal lattices)**.** A *lattice* in $\mathbb{R}^n$ is a subgroup $\Gamma = A\mathbb{Z}^n$ for some invertible linear map $A : \mathbb{R}^n \to \mathbb{R}^n$. Its covolume is $\mathrm{vol}(\Gamma) = |\det A|$, and a fundamental cell is $F = A[0,1)^n$. Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

Exercise 2.19 (Fourier modes of a lattice)**.** Prove $\Gamma^* = 2\pi(A^{-1})^*\mathbb{Z}^n$. Show that $e_\xi(x) = e^{i\xi\cdot x}$ is Γ-periodic exactly when $\xi \in \Gamma^*$. With inner product

$$\langle f, g\rangle_F = \frac{1}{\mathrm{vol}(\Gamma)} \int_F \overline{f(x)} g(x)\, dx,$$

prove that $(e_\xi)_{\xi\in\Gamma^*}$ is a complete orthonormal family in $L^2(\mathbb{R}^n/\Gamma)$. Interpret this as $\widehat{\mathbb{R}^n/\Gamma} \cong \Gamma^*$.

Definition 2.15 (Periodic summation and structure factors)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, its Γ-*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma\in\Gamma} f(x+\gamma).$$

For a finite motif $B \subset \mathbb{R}^n$ with scattering weights c_b, its *structure factor* is

$$S_B(q) = \sum_{b\in B} c_b e^{-iq\cdot b}.$$

Exercise 2.20 (Poisson summation)**.** Prove that $P_\Gamma f$ converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\mathrm{vol}(\Gamma)} \sum_{\xi\in\Gamma^*} \mathcal{F}f(\xi) e^{i\xi\cdot x}.$$

Set $x = 0$ to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

Exercise 2.21 (Crystal diffraction and Bragg peaks)**.** Let Γ_N be a finite rectangular portion of Γ. In the single-scattering approximation, prove that the amplitude at momentum transfer q factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq\cdot\gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at $q \in \Gamma^*$ as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude $2\pi/\lambda$, derive the Laue condition and, for planes spaced by d, the Bragg relation

$$2d \sin\theta = m\lambda.$$

Show how zeros of S_B produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

Exercise 2.22 (Electron diffraction)**.** For an electron accelerated through voltage V, use $eV = p^2/(2m_e)$ and $p = h/\lambda$ to derive

$$\lambda = \frac{h}{\sqrt{2m_e eV}}.$$

Combine this relation with the reciprocal-lattice and Bragg conditions to predict how the diffraction angles change with V. Compare the predicted sharply directed beams with Davisson–Germer, *Physical Review* **30** (1927).

Definition 2.16 (Compact-group Fourier transform)**.** For a compact group G, let $\widehat{G}$ denote the equivalence classes of irreducible finite-dimensional unitary representations. For $f \in L^1(G)$ and $\pi \in \widehat{G}$, define

$$\widehat{f}(\pi) = \int_G f(g)\pi(g)^*\, dg \in \operatorname{End}(V_\pi).$$

Thus the Fourier transform is scalar-valued for abelian G and matrix-valued in general.

Exercise 2.23 (Peter–Weyl theorem)**.** Use Haar averaging and Schur's lemma to prove orthogonality of matrix coefficients of irreducible unitary representations of a compact group. Prove that their span is dense in $C(G)$ and $L^2(G)$ and obtain

$$L^2(G) \cong \widehat{\bigoplus_{\pi\in\widehat{G}}} V_\pi \otimes V_\pi^*.$$

Derive Fourier inversion, the Plancherel formula

$$\|f\|_2^2 = \sum_{\pi\in\widehat{G}} (\dim V_\pi)\|\widehat{f}(\pi)\|_{\mathrm{HS}}^2,$$

and $\widehat{f * g}(\pi) = \widehat{g}(\pi)\widehat{f}(\pi)$. Recover ordinary Fourier series for $G = \mathbb{T}$. For $G = SO(3)$, identify the matrix coefficients with the Wigner functions D^ℓ_{mn}.

Exercise 2.24 (Spherical harmonics and atomic orbitals)**.** Identify S^2 with $SO(3)/SO(2)$ and $L^2(S^2)$ with the right $SO(2)$-invariant subspace of $L^2(SO(3))$. Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell$$

and identify a basis of V_ℓ with $Y_{\ell m} \propto D^{\ell}_{m0}$. Derive $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1) Y_{\ell m}$ and completeness of the spherical harmonics. Separate the Coulomb Hamiltonian in spherical coordinates to obtain $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r) Y_{\ell m}(\theta,\phi)$. Determine the angular nodes of the s, p, and d orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare with Matsui et al., *Physical Review B* **72** (2005).

Exercise 2.25 (Molecular rotational spectra)**.** Model a rigid molecule with orientation $R \in SO(3)$ and principal moments I_1, I_2, I_3. On $L^2(SO(3))$, derive

$$H_{\mathrm{rot}} = \frac{J_1^2}{2I_1} + \frac{J_2^2}{2I_2} + \frac{J_3^2}{2I_3}.$$

Use the Peter–Weyl basis to reduce H_{rot} to finite-dimensional blocks. For a symmetric top, derive

$$E_{JK} = \frac{\hbar^2 J(J+1)}{2I_\perp} + \frac{\hbar^2 K^2}{2}\left(\frac{1}{I_\parallel} - \frac{1}{I_\perp}\right).$$

Derive the electric-dipole selection rules and the splitting in a static electric field. Compare the predicted components with microwave measurements of the symmetric-top molecule s-trioxane reported in NBS Special Publication 343 (1971).

Definition 2.17 (Noncommutative Plancherel transform)**.** A locally compact group is *unimodular* when its left and right Haar measures agree. It is *type I* when its unitary representations have essentially unique direct-integral decompositions into irreducibles. Let G be second-countable, unimodular, and type I. For $f \in L^1(G)$ and an irreducible unitary representation $\pi : G \to U(H_\pi)$, define the operator-valued Fourier transform

$$\widehat{f}(\pi) = \int_G f(g)\pi(g)^*\, dg.$$

The noncommutative Plancherel theorem supplies a measure ν_G on the unitary dual $\widehat{G}$ such that, initially for $f \in L^1(G) \cap L^2(G)$,

$$\|f\|_2^2 = \int_{\widehat{G}} \|\widehat{f}(\pi)\|_{\mathrm{HS}}^2\, d\nu_G(\pi), \qquad \widehat{f * g}(\pi) = \widehat{g}(\pi)\widehat{f}(\pi).$$

For abelian groups the representation spaces are one-dimensional; for compact groups the integral becomes the dimension-weighted Peter–Weyl sum.

Definition 2.18 (Heisenberg group and Schrödinger representations)**.** On $\mathbb{R}^{2n} = \mathbb{R}^n_q \oplus \mathbb{R}^n_p$, set

$$\sigma((q,p),(q',p')) = p \cdot q' - q \cdot p'.$$

The *Heisenberg group* is $\mathbb{H}_n = \mathbb{R}^{2n} \times \mathbb{R}$ with multiplication

$$(z,s)(z',s') = (z+z', s+s'+\tfrac{1}{2}\sigma(z,z')).$$

For $\lambda \neq 0$, its *Schrödinger representation* on $L^2(\mathbb{R}^n)$ is

$$(\pi_\lambda(q,p,s)\psi)(x) = e^{i\lambda(s+p\cdot(x-q/2))}\psi(x-q).$$

The physical central character is $\lambda = 1/\hbar$.

Exercise 2.26 (Harmonic analysis of the Heisenberg group)**.** Verify the group law, identify the center, and prove that Lebesgue measure is Haar measure on $\mathbb{H}_n$. Prove that each π_λ is an irreducible unitary representation and derive its Weyl commutation relations. At $\lambda = 1/\hbar$, differentiate the position and momentum subgroups and recover Q_j, P_j, and $[Q_j, P_k] = i\hbar\delta_{jk}$. Taking the Stone–von Neumann theorem as input, show that every irreducible representation with central character $e^{i\lambda s}$ is unitarily equivalent to π_λ.

Compute the operator-valued group Fourier transform in these representations and derive the Plancherel decomposition with measure $c_n|\lambda|^n \, d\lambda$, determining c_n from the Haar and Fourier normalizations above. Show that the one-dimensional representations with trivial central character have zero Plancherel measure. Derive the distributional orthogonality relation for Weyl operators and the resulting Hilbert–Schmidt Plancherel and inversion formulas. Prove that

$$S \longmapsto \chi_S(u, v) = \mathrm{Tr}\Big(Se^{i(u\cdot X+v\cdot Y)}\Big)$$

is injective on trace-class operators, where $[X_j, Y_k] = i\delta_{jk}$. Interpret this as noncommutative Fourier analysis turning functions on a group into operators rather than scalar frequency amplitudes.

Part III

Fields

3 Geometry and Dynamics of Fields

Experimental observations.

- A changing magnetic flux through a conducting loop produces an electromotive force proportional to the rate of flux change.
- GW150914 produced a peak interferometer strain of order 10^{-21} near 150 Hz in two widely separated detectors.

3.1 Vector Fields and Differential Forms

Definition 3.1 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. A *flow* of X is a smooth local action Φ of $\mathbb{R}$ on M satisfying $\frac{d}{dt}\Big|_0 \Phi_t(p) = X_p$. The *Lie bracket* is the vector field $[X, Y](f) = X(Yf) - Y(Xf)$.

Exercise 3.1 (Flows and brackets)**.** Prove local existence and uniqueness of flows. Prove that $[X, Y] = 0$ if and only if their local flows commute, and that the Lie derivative $\mathcal{L}_X Y = [X, Y]$ is natural under diffeomorphisms.

Definition 3.2 (Differential forms and pullback)**.** A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$, using the exterior powers defined above; write $\Omega^k(M) = \Gamma(\Lambda^k T^*M)$. The *pullback* of $\omega \in \Omega^k(N)$ by $F : M \to N$ is

$$(F^*\omega)_p(v_1, \ldots, v_k) = \omega_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

Definition 3.3 (Exterior derivative)**.** The *exterior derivative* is the unique family of maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and naturality $F^*d = dF^*$.

Exercise 3.2 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity $\mathcal{L}_X = d\iota_X + \iota_X d$ and the naturality of d, wedge products, and pullbacks.

Definition 3.4 (Integration of forms)**.** An *orientation* of an n-manifold is a continuous choice of component of nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 3.3 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form B and electric one-form E, prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt}\int_S B \qquad \text{and} \qquad dE = -\partial_t B,$$

recovering the observed flux–emf law.

3.2 Metrics and Spacetime

Differential forms describe fluxes without a metric. A metric adds lengths, proper times, volume, and the duality between complementary form degrees.

Definition 3.5 (Tensor fields and metrics)**.** A (r, s)-*tensor field* is a smooth section of $TM^{\otimes r} \otimes T^*M^{\otimes s}$. A *Riemannian metric* is a smooth positive-definite inner product on each T_pM. A *Lorentzian metric* is a smooth nondegenerate symmetric form of signature $(-, +, \dots, +)$. The length of a curve in a Riemannian manifold is $\int \sqrt{g(\dot\gamma, \dot\gamma)}\, dt$; the proper time of a timelike curve is $c^{-1} \int \sqrt{-g(\dot\gamma, \dot\gamma)}\, dt$. A metric and an orientation define a volume form vol_g.

Definition 3.6 (Hodge star)**.** On an oriented pseudo-Riemannian n-manifold, the metric induces a nondegenerate pairing $\langle \cdot, \cdot \rangle_g$ on differential forms. The *Hodge star* is the bundle map

$$* : \Omega^p(M) \longrightarrow \Omega^{n-p}(M)$$

characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle_g \ \mathrm{vol}_g$$

for p-forms α, β.

Exercise 3.4 (Metric volume and Hodge star)**.** Construct vol_g and the Hodge star intrinsically and derive their coordinate expressions. For a metric with q negative directions, prove on p-forms that

$$*^2 = (-1)^{p(n-p)+q}\, \mathrm{id}\,.$$

Compute $*$ on the standard coordinate forms of Euclidean three-space and Minkowski four-space, stating the orientation and signature conventions.

Exercise 3.5 (Proper time)**.** Prove invariance of length and proper time under reparametrization. In Minkowski spacetime, prove that the inertial path maximizes proper time among piecewise smooth future-directed timelike paths with fixed timelike-separated endpoints, and derive special-relativistic time dilation. Using the muon rest lifetime $\tau_\mu \simeq 2.2\,\mu$s, derive the mean laboratory travel distance $\gamma v \tau_\mu$ and compare it with the distance $v\tau_\mu$ predicted without time dilation.

3.3 Connections, Curvature, and Gauge Fields

A connection compares fields at different points. Its curvature records the failure of parallel transport to be path independent and supplies the field strength for both geometry and gauge theory.

Definition 3.7 (Connections and geodesics)**.** A *connection* on TM is an $\mathbb{R}$-bilinear map $\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fY) = X(f)Y + f\nabla_X Y$. Its torsion is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$. A *geodesic* satisfies $\nabla_{\dot\gamma}\dot\gamma = 0$. A connection is *metric compatible* if $Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

Exercise 3.6 (Levi–Civita connection)**.** Prove that every Riemannian or Lorentzian metric has a unique torsion-free, metric-compatible connection. Derive the Koszul formula and prove that its geodesics are precisely the critical curves of the energy functional with fixed endpoints.

Definition 3.8 (Covariant derivative and parallel transport)**.** For a curve γ, the connection induces a *covariant derivative* D/dt on vector fields along γ. A field V is *parallel* when $DV/dt = 0$; the resulting linear isomorphism between endpoint tangent spaces is *parallel transport*.

Exercise 3.7 (Parallel transport and holonomy)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Show that Levi–Civita transport preserves the metric. Compute the rotation obtained by transporting a tangent vector around a latitude on the unit sphere and relate it to the enclosed curvature.

Definition 3.9 (Curvature)**.** The *curvature* of a connection is

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For a metric connection, the *Riemann tensor* is $\mathrm{Riem}(X,Y,Z,W) = g(R(X,Y)Z, W)$. Its contraction is the *Ricci tensor* Ric; the contraction of Ricci is the *scalar curvature* S.

Exercise 3.8 (Curvature identities and geodesic deviation)**.** For the Levi–Civita connection, prove tensoriality and the algebraic symmetries of the Riemann tensor and the first and second Bianchi identities. For a variation through geodesics with separation field J, prove

$$\frac{D^2 J}{dt^2} + R(J,\dot\gamma)\dot\gamma = 0.$$

Deduce the leading relative acceleration of nearby freely falling bodies.

Definition 3.10 (Smooth bundles and sections)**.** A *smooth fiber bundle* with fiber F is a smooth map $\pi : E \to M$ locally isomorphic over M to $U \times F \to U$. A *smooth vector bundle* has vector-space fibers and linear transition maps. A *smooth principal G-bundle* has a free right G-action, transitive on each fiber, and local equivariant trivializations. A *section* is a smooth map $s : M \to E$ with $\pi s = \mathrm{id}_M$. For a representation $G \to GL(V)$, the *associated vector bundle* is $P \times_G V = (P \times V)/((pg, v) \sim (p, gv))$.

Definition 3.11 (Gauge fields)**.** Let G be a Lie group and $\pi : P \to M$ a principal G-bundle. A *gauge connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. Its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M.

Exercise 3.9 (Gauge covariance and Bianchi identity)**.** Choose local sections s_i and derive transition maps $g_{ij} : U_i \cap U_j \to G$ and the transformation laws for the local potentials $A_i = s_i^* A$ and curvatures $F_i = s_i^* F_A$. Prove $d_A F_A = 0$ and gauge invariance of holonomy up to conjugacy. For $G = U(1)$, recover $A \mapsto A + d\chi$ and $F = dA$. On $\mathbb{R}^2 \setminus \{0\}$, compute the curvature and holonomy of $A = (\Phi/2\pi)d\theta$ and derive the Aharonov–Bohm phase shift $q\Phi/\hbar$ despite $F = 0$ along the paths.

Exercise 3.10 (Yang–Mills equations and instantons)**.** For $G = SU(r)$, use $\langle X, Y\rangle = -2\,\mathrm{Tr}(XY)$ and define

$$S_{\mathrm{YM}}(A) = \frac{1}{2g^2}\int_M \langle F_A \wedge *F_A\rangle$$

on a closed oriented Riemannian four-manifold. Prove gauge invariance and derive $d_A^* F_A = 0$, where $d_A^* = - * d_A *$. Decompose $F_A = F_A^+ + F_A^-$ and, for

$$k(A) = -\frac{1}{8\pi^2}\int_M \mathrm{Tr}(F_A \wedge F_A),$$

prove $S_{\mathrm{YM}}(A) \geq 8\pi^2|k(A)|/g^2$, with equality precisely for $F_A = \pm * F_A$. Use the Bianchi identity to show that these instantons solve the Yang–Mills equation.

3.4 Classical Field Equations

The geometric language now expresses electromagnetism and gravitation as equations for curvature constrained by conservation laws.

Definition 3.12 (Electromagnetic field)**.** On an oriented Lorentzian four-manifold, an *electromagnetic field strength* is a two-form F. Given a current one-form J and constants μ_0, c, *Maxwell's equations* are

$$dF = 0, \qquad d{*}F = \mu_0 {*}J.$$

Locally, a *gauge potential* is a one-form A with $F = dA$.

Exercise 3.11 (Maxwell theory and electromagnetic waves)**.** Derive charge conservation from Maxwell's equations. On Minkowski spacetime, recover the electric and magnetic vector equations and the Lorentz force from F. In vacuum, impose Lorenz gauge and derive wave equations with speed c. Prove transverse polarization and compare the predicted relation $c = (\mu_0\varepsilon_0)^{-1/2}$ with the measured speed of light.

Definition 3.13 (Einstein equation)**.** The *Einstein tensor* of a Lorentzian metric is $G = \mathrm{Ric} - \frac{1}{2}Sg$. A *stress–energy tensor* is a symmetric two-tensor T whose contraction with an observer's velocity gives the observer's energy–momentum current. The *Einstein equation* with cosmological constant Λ is

$$G + \Lambda g = \frac{8\pi G_N}{c^4}T.$$

Exercise 3.12 (Conservation and the Newtonian limit)**.** Use the contracted Bianchi identity to derive $\nabla^a T_{ab} = 0$. Linearize $g = \eta + h$ for a static weak field and recover Poisson's equation and the Newtonian inverse-square law with the stated coupling constant.

Exercise 3.13 (Gravitational radiation)**.** Linearize the vacuum Einstein equation about Minkowski spacetime, impose transverse–traceless gauge, and obtain two tensor polarizations satisfying a wave equation. Apply geodesic deviation to a ring of freely falling test masses and derive the strain measured by an orthogonal-arm interferometer.

Exercise 3.14 (Gravitational-wave detection)**.** For a quasi-circular binary, derive the leading quadrupole waveform and the chirp relation between frequency, frequency derivative, and chirp mass. For GW150914, use the reported detector-frame component-mass medians $36M_\odot$ and $29M_\odot$ to infer the chirp mass. Compare with the model-independent reconstruction $\mathcal{M} \simeq 30M_\odot$ and with the reported peak strain $h \simeq 10^{-21}$ near $150\,\mathrm{Hz}$ in Abbott et al., *Physical Review D* **93** (2016) and the LIGO GW150914 fact sheet. Limit the conclusion to agreement of the observed waveform with general relativity in that regime.

Part IV

States

4 Classical, Statistical, and Quantum States

Experimental observations.

- A dilute gas obeys the ideal-gas law and exhibits a temperature-dependent distribution of molecular speeds.
- Repeated Stern–Gerlach measurements yield two outcomes whose relative frequencies depend on the preparation and analyzer directions.

4.1 Classical States and Hamiltonian Dynamics

Definition 4.1 (Configuration and phase spaces)**.** A *configuration space* is a smooth manifold Q. Its *phase space* is the cotangent bundle T^*Q. The tautological one-form on T^*Q is $\theta_\alpha(v) = \alpha(d\pi(v))$, and its canonical symplectic form is $\omega = -d\theta$.

Definition 4.2 (Classical states and observables)**.** On a phase space M, a *pure classical state* is a point $x \in M$, and a *classical observable* is a smooth real-valued function $f \in C^\infty(M, \mathbb{R})$. A *statistical state* is a Borel probability measure μ on M, with expectation

$$\mathbb{E}_\mu[f] = \int_M f\, d\mu$$

whenever the integral exists. If ϕ_t is a Hamiltonian flow, Schrödinger evolution pushes states forward by $\mu_t = (\phi_t)_*\mu_0$, while Heisenberg evolution pulls observables back by $f_t = f \circ \phi_t$; hence $\int_M f\, d\mu_t = \int_M f_t\, d\mu_0$.

Definition 4.3 (Symplectic dynamics)**.** A *symplectic form* is a closed, nondegenerate two-form ω. The *Hamiltonian vector field* of $H \in C^\infty(M)$ is defined by $\iota_{X_H}\omega = dH$. Its flow is the *Hamiltonian flow.* The *Poisson bracket* is $\{f, g\} = \omega(X_f, X_g)$.

Exercise 4.1 (Hamilton's equations and Poisson algebra)**.** Prove that the canonical form on T^*Q is symplectic. In canonical coordinates, derive Hamilton's equations. Prove antisymmetry, the Leibniz rule, and the Jacobi identity for the Poisson bracket, and prove $\dot{f} = \{f, H\} + \partial_t f$ along Hamiltonian flow.

Definition 4.4 (Symplectic actions and momentum maps)**.** An action of a Lie group G on (M, ω) is *symplectic* if it preserves ω. A *momentum map* is an equivariant map $J : M \to \mathfrak{g}^*$ such that $d\langle J, \xi\rangle = \iota_{\xi_M}\omega$ for every $\xi \in \mathfrak{g}$.

Definition 4.5 (Symplectic reduction)**.** Let $J : M \to \mathfrak{g}^*$ be a momentum map, let ξ be a regular value, and let $G_\xi = \{g \in G : \mathrm{Ad}_g^*\,\xi = \xi\}$. If G_ξ acts freely and properly on $J^{-1}(\xi)$, the *reduced phase space* is

$$M//_\xi G = J^{-1}(\xi)/G_\xi.$$

Writing $i : J^{-1}(\xi) \hookrightarrow M$ and $q : J^{-1}(\xi) \to M//_\xi G$, its *reduced symplectic form* is the form ω_ξ characterized by $q^*\omega_\xi = i^*\omega$.

Exercise 4.2 (Noether's theorem)**.** Prove that if a Hamiltonian is invariant under a symplectic G-action with momentum map J, then every component of J is conserved. Compute the momentum maps for translations and rotations of a particle and recover linear and angular momentum conservation.

Exercise 4.3 (Reduction and central-force dynamics)**.** Under the hypotheses of symplectic reduction, prove that $i^*\omega$ is basic for the G_ξ-action and that ω_ξ exists uniquely and is symplectic. For a G-invariant Hamiltonian H, prove that X_H is tangent to $J^{-1}(\xi)$, that H descends to a Hamiltonian H_ξ, and that the flow of H_ξ is the quotient of the flow of H.

Apply this to the cotangent-lifted rotation action on $T^*(\mathbb{R}^3 \setminus \{0\})$, whose momentum map is $J(q,p) = q \times p$. For a nonzero angular momentum ℓ, identify $J^{-1}(\ell)/SO(3)_\ell$ with the radial phase space having coordinates (r, p_r), prove that its symplectic form is $dr \wedge dp_r$, and reduce

$$H(q,p) = \frac{\|p\|^2}{2m} + V(\|q\|)$$

to

$$H_\ell(r, p_r) = \frac{p_r^2}{2m} + V(r) + \frac{\|\ell\|^2}{2mr^2}.$$

For $V(r) = -\kappa/r$, derive the elliptical bound orbits and the equal-area law.

Exercise 4.4 (Gauss constraints and transverse modes)**.** Let $d : V \to W$ be a linear map between finite-dimensional real inner-product spaces. On $M = W \oplus W$ set

$$\omega((a,e),(a',e')) = \langle a, e'\rangle - \langle a', e\rangle.$$

Let the additive group $(\ker d)^\perp$ act by $\chi \cdot (A,E) = (A + d\chi, E)$. Prove that this action is free and Hamiltonian with momentum map

$$\langle J(A,E), \chi\rangle = \langle d\chi, E\rangle = \langle \chi, d^*E\rangle.$$

Show that reduction at zero imposes $d^*E = 0$ and identifies

$$M//_0(\ker d)^\perp \cong (W/\operatorname{im} d) \oplus \ker d^* \cong \ker d^* \oplus \ker d^*.$$

Interpret d as a finite-dimensional approximation to the exterior derivative: $d^*E = 0$ is the vacuum Gauss constraint, $A \sim A + d\chi$ is gauge equivalence, and the reduced variables are the transverse electromagnetic modes.

Exercise 4.5 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the phase-space volume $\omega^n/n!$, and every density depending only on H. Deduce that Hamiltonian evolution cannot compress phase volume.

Exercise 4.6 (Betatron phase portraits)**.** Model one transverse degree of freedom in a storage ring by $H = (p^2 + \omega^2 q^2)/2$. Derive the closed elliptical phase portraits and show that their enclosed area is preserved. Interpret the motion as a betatron oscillation about a closed orbit.

4.2 Probability, Gibbs States, and Ergodicity

Probability measures enlarge pure classical states to statistical ensembles. Entropy and invariance then distinguish equilibrium among those ensembles.

Definition 4.6 (Probability). A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A *random variable* is a measurable map. Its *expectation* is its integral. For $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, a *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is a $\mathcal{G}$-measurable integrable random variable satisfying $\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P}$ for all $G \in \mathcal{G}$. Families of σ-algebras are *independent* when the probability of every finite intersection is the product of its probabilities.

Exercise 4.7 (Radon–Nikodym). For finite measures $\nu \ll \mu$, prove that $\nu(A) = \int_A f\, d\mu$ for a unique $f \in L^1(\mu)$, $f \geq 0$. Extend the statement to finite signed measures.

Exercise 4.8 (Expectation, conditioning, and independence). Use Radon–Nikodym to prove existence and almost-everywhere uniqueness of conditional expectation. Prove the tower property and conditional Jensen inequality. Characterize independence by factorization of expectations and derive variance addition for independent square-integrable random variables.

Definition 4.7 (Entropy and Gibbs ensembles). For a discrete probability vector p, its *entropy* is $S(p) = -k_B \sum_i p_i \log p_i$. For a phase-space probability density ρ relative to Liouville measure μ_L, its *statistical entropy* is $S(\rho) = -k_B \int \rho \log \rho\, d\mu_L$. The *canonical Gibbs ensemble* at inverse temperature β is defined when

$$0 < Z(\beta) = \int e^{-\beta H}\, d\mu_L < \infty,$$

and is then the probability measure $d\mathbb{P}_\beta = Z(\beta)^{-1} e^{-\beta H} d\mu_L$.

Definition 4.8 (Measure-preserving dynamics and ergodicity). A measurable transformation T is *measure preserving* if $\mu(T^{-1}A) = \mu(A)$. It is *ergodic* if every invariant measurable set has measure zero or full measure. An *equilibrium state* is an invariant probability measure for the dynamics.

Exercise 4.9 (Entropy and equilibrium). Maximize entropy under normalization and fixed mean energy to derive the Gibbs ensemble. Prove $\langle H \rangle = -\partial_\beta \log Z$ and $\mathrm{Var}(H) = \partial_\beta^2 \log Z$. Prove invariance of Gibbs measure under Hamiltonian flow and formulate the consequence of the ergodic theorem for equality of time and ensemble averages.

Exercise 4.10 (Ideal-gas thermodynamics). Compute the classical canonical partition function of N indistinguishable noninteracting particles in a box, including the phase-space normalization $h^{3N} N!$. Derive the ideal-gas law, internal energy, and entropy, and the one-particle Maxwell–Boltzmann speed distribution. Derive its temperature scaling and compare both predictions with dilute-gas measurements in the classical regime.

4.3 Quantum States and Quantization

Quantum states retain the probabilistic expectation rule while replacing commutative observables by operators and classical mixtures by density operators.

Definition 4.9 (Quantum states and observables). A positive operator S has the coordinate-free trace

$$\mathrm{Tr}\, S = \sup_P \mathrm{tr}\left((PSP)|_{PH}\right),$$

where P ranges over orthogonal projections with finite-dimensional image. A bounded operator T is *trace class* when $\mathrm{Tr}\,|T| < \infty$, where $|T| = (T^*T)^{1/2}$; its trace is obtained by linearity from the positive and negative parts of its real and imaginary parts. A *density operator* is positive and trace class with $\operatorname{Tr}\rho = 1$. An *observable* is a self-adjoint operator A with spectral measure E_A. The *Born probability* of an event B is $\operatorname{Tr}(\rho E_A(B))$. A closed-system *unitary evolution* is $\rho(t) = U_t \rho(0) U_t^*$, where $(U_t)_{t\in\mathbb{R}}$ is a strongly continuous one-parameter unitary group. For unbounded observables and generators, spectral measures, expectations, and exponentials use the earlier unbounded spectral calculus and its natural domains.

Exercise 4.11 (Classical and quantum state spaces)**.** Let X be compact Hausdorff and let H be finite-dimensional. Show that a point $x \in X$, a probability measure μ on X, a unit vector $\psi \in H$, and a density operator ρ determine states by

$$\delta_x(f) = f(x), \qquad \omega_\mu(f) = \int_X f\, d\mu, \qquad \omega_\psi(A) = \langle \psi, A\psi\rangle, \qquad \omega_\rho(A) = \operatorname{Tr}(\rho A).$$

Prove that the extreme points of the classical state space are the point evaluations and that the extreme points of the quantum state space are the rank-one density operators. Explain why mixtures are convex combinations in both theories and why the structural distinction is that $C(X)$ is commutative whereas $B(H)$ generally is not.

Exercise 4.12 (Projective quantum states by reduction)**.** Let K be a finite-dimensional complex Hilbert space and regard its underlying real vector space as symplectic with

$$\omega(u, v) = -\operatorname{Im}\langle u, v\rangle.$$

For the scalar $U(1)$-action, identify its Lie algebra with $\mathbb{R}$ acting infinitesimally by $z \mapsto iz$, and prove that $J(z) = \|z\|^2/2$ is a momentum map. Show that reduction at $1/2$ gives

$$K//_{1/2}U(1) = S(K)/U(1) = \mathbb{P}(K),$$

the space of complex lines in K. Identify $[z] \in \mathbb{P}(K)$ with the rank-one density operator

$$P_z : v \longmapsto z\langle z, v\rangle$$

for a unit representative z, and hence with the pure state $A \mapsto \operatorname{Tr}(P_z A)$. Prove that the reduced symplectic form is the Fubini–Study form, characterized on horizontal representatives $u, v \in z^\perp$ by

$$(\omega_{\mathrm{FS}})_{[z]}([u], [v]) = -\operatorname{Im}\langle u, v\rangle.$$

Thus normalization and quotient by global phase produce the pure quantum state space as a symplectic reduction.

Exercise 4.13 (Canonical quantization and Weyl relations)**.** On $\mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$, set $Q_j\psi = x_j\psi$ and $P_j\psi = -i\hbar\partial_j\psi$. Prove $[Q_j, P_k] = i\hbar\delta_{jk}$ and derive the Weyl relations for their unitary groups. Using symmetric ordering, verify the Poisson-bracket–commutator rule for affine and quadratic observables. Compare different orderings of q^2p^2 and explain the Groenewold–van Hove obstruction to quantizing all observables by this rule.

Exercise 4.14 (Quantized harmonic oscillator)**.** For $H = P^2/(2m) + m\omega^2Q^2/2$, define creation and annihilation operators on the Schwartz core $\mathcal{S}(\mathbb{R})$ and prove there that $[a, a^*] = 1$ and $H = \hbar\omega(a^*a + 1/2)$. Construct the eigenstates with $E_n = \hbar\omega(n + 1/2)$ and use the spectral-lines exercise to obtain the frequency spacing ω. Prove that their span is a core for the self-adjoint closure of H. For a two-level internal transition of frequency Ω_0 coupled linearly to Q, compute the oscillator matrix elements, obtain motional sidebands at $\Omega_0 \pm \omega$, and compare their spacing with trapped-ion spectra.

Definition 4.10 (Quadratures, Wigner distributions, and homodyne data)**.** For one canonical degree of freedom, choose $q_0, p_0 > 0$ with $q_0 p_0 = \hbar$ and set

$$X = Q/q_0, \qquad Y = P/p_0, \qquad X_\theta = X\cos\theta + Y\sin\theta,$$

so that $[X, Y] = i$. For a density operator ρ, its *Weyl characteristic function* is

$$\chi_\rho(u, v) = \mathrm{Tr}\Big(\rho e^{i(uX+vY)}\Big),$$

and its *Wigner distribution* is the inverse Fourier transform

$$W_\rho(x, y) = \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \chi_\rho(u, v) e^{-i(ux+vy)}\, du\, dv,$$

interpreted as a tempered distribution when necessary. If E_θ is the spectral measure of X_θ, the *homodyne quadrature distribution* is

$$p_\rho(r \mid \theta)\, dr = \mathrm{Tr}\big(\rho E_\theta(dr)\big).$$

In balanced optical homodyne detection, the local-oscillator phase selects θ and repeated measurements estimate this distribution.

Exercise 4.15 (Optical homodyne tomography)**.** Use Exercise 2.26 and the spectral theorem to prove the Fourier-slice identity

$$\widetilde{p}_\rho(k \mid \theta) = \int_{\mathbb{R}} p_\rho(r \mid \theta) e^{ikr}\, dr = \chi_\rho(k\cos\theta, k\sin\theta).$$

Deduce, in the distributional sense, that the measured quadrature density is the Radon transform of the Wigner distribution:

$$p_\rho(r \mid \theta) = \int_{\mathbb{R}^2} W_\rho(x, y)\delta(r - x\cos\theta - y\sin\theta)\, dx\, dy.$$

Prove that the data for $0 \le \theta < \pi$ determine χ_ρ, hence W_ρ and ρ, and derive the filtered-backprojection formula

$$W_\rho(x, y) = \frac{1}{(2\pi)^2}\int_0^\pi \int_{\mathbb{R}} |k|\, \widetilde{p}_\rho(k \mid \theta) e^{-ik(x\cos\theta + y\sin\theta)}\, dk\, d\theta.$$

Show that every $p_\rho(\,\cdot \mid \theta)$ is a genuine probability density although W_ρ may be negative. Compute the data and reconstruction for vacuum, coherent, squeezed, and one-photon states. Explain why the factor $|k|$ amplifies high-frequency noise and why finite-angle experimental data require regularization or constrained density-matrix estimation. Compare the predicted reconstruction of vacuum and squeezed states with Smithey et al., *Physical Review Letters* **70** (1993).

Exercise 4.16 (Liouville and von Neumann dynamics)**.** Prove that the Born rule defines a probability measure and that its expected value is $\mathrm{Tr}(\rho A)$ when defined. Derive the following classical and quantum evolution equations.

Let $\mu_t = (\phi_t)_*\mu_0$ have smooth density p_t relative to the Liouville measure μ_L of a time-independent Hamiltonian system. Using Exercise 4.5, prove

$$\partial_t p_t + \{p_t, H\} = 0, \qquad \frac{d}{dt}\int_M f\, d\mu_t = \int_M \{f, H\}\, d\mu_t.$$

For $U_t = e^{-itH/\hbar}$, set $\rho_t = U_t \rho_0 U_t^*$ and $A_t = U_t^* A_0 U_t$. Prove

$$\dot{\rho}_t = -\frac{i}{\hbar}[H, \rho_t], \qquad \dot{A}_t = \frac{i}{\hbar}[H, A_t].$$

Prove

$$\int_M f\, d\mu_t = \int_M (f \circ \phi_t)\, d\mu_0, \qquad \mathrm{Tr}(\rho_t A_0) = \mathrm{Tr}(\rho_0 A_t),$$

and identify these as the equivalence of Schrödinger and Heisenberg pictures. Compare the derivations $f \mapsto \{f, H\}$ and $A \mapsto (i\hbar)^{-1}[A, H]$, and prove conservation of expected energy in both theories.

Exercise 4.17 (Stern–Gerlach projections and the Born rule)**.** For a spin-1/2 system and unit vector n, use $(n \cdot \sigma)^2 = \mathrm{id}$ to construct the complete spectral projections of $S_n = (\hbar/2) n \cdot \sigma$ as $P_\pm(n) = (1 \pm n \cdot \sigma)/2$. For $\rho = (1 + r \cdot \sigma)/2$, derive $p_\pm(n) = \mathrm{Tr}(\rho P_\pm(n)) = (1 \pm r \cdot n)/2$. Model an inhomogeneous magnetic field by $H_{\mathrm{int}} = -\boldsymbol{\mu} \cdot \mathbf{B}$ and explain the separation into discrete beams. For successive analyzers along axes n and m, derive the transition probability $\mathrm{Tr}(P_+(n)P_+(m)) = (1 + n \cdot m)/2$ and interpret the intermediate projection. Compare the two-way splitting with Gerlach–Stern, *Zeitschrift für Physik* **9** (1922). Distinguish the original observation of space quantization in silver atoms from its later spin interpretation, and state which discrete outcomes and frequency predictions test the projection postulate and Born rule.

Exercise 4.18 (Blackbody radiation from quantized Fourier modes)**.** For electromagnetic radiation in a large conducting cavity, use its Fourier modes and two polarizations to derive the mode density $g(\omega) = \omega^2/(\pi^2 c^3)$ per unit volume. Show that classical Gibbs equipartition assigns mean energy $k_B T$ to each mode and produces the ultraviolet divergence. Using the oscillator levels derived in Exercise 4.14, compute the quantum Gibbs partition function and the thermal mean energy $\hbar\omega/(e^{\beta\hbar\omega} - 1)$. Deduce Planck's law

$$u(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/(k_B T)} - 1},$$

the Stefan–Boltzmann T^4 law, and Wien scaling $\omega_{\max} \propto T$. Compare the low- and high-frequency limits with the cavity spectra discussed by Planck, *Annalen der Physik* **309** (1901), and identify precisely where quantization removes the classical divergence.

Definition 4.11 (C^*-algebras and states)**.** A *unital Banach algebra* is a complete normed algebra with identity and $\|ab\| \le \|a\|\|b\|$. An *involution* satisfies $(ab)^* = b^* a^*$ and $(a^*)^* = a$. A *C^*-algebra* is an involutive Banach algebra satisfying $\|a^* a\| = \|a\|^2$. An element is *positive* if it is $b^* b$ for some b. A *state* is a positive linear functional φ with $\varphi(1) = 1$; positivity then implies $\|\varphi\| = 1$.

Exercise 4.19 (Commutative Gelfand duality)**.** For compact Hausdorff X, prove that $C(X)$ is a commutative unital C^*-algebra. Let $\operatorname{Spec} A$ be the weak-$*$ compact space of nonzero unital $*$-homomorphisms $A \to \mathbb{C}$. Prove that every commutative unital C^*-algebra A is isometrically $*$-isomorphic to $C(\operatorname{Spec} A)$ by the Gelfand transform, and identify states on $C(X)$ with probability measures.

5 Composite and Many-Body States

Experimental observations.

- Tomography of prepared entangled pairs reconstructs a nearly pure joint state while either subsystem alone has nearly equal two-outcome frequencies.

Definition 5.1 (Composite and reduced states)**.** The Hilbert space of a composite system is $H_A \otimes H_B$, and its normal states are the density operators on this tensor product. A density operator is *separable* if it lies in the trace-norm closed convex hull of product density operators and *entangled* otherwise. The *reduced state* $\rho_A = \operatorname{Tr}_B \rho$ is determined by $\operatorname{Tr}(\rho_A A) = \operatorname{Tr}(\rho(A \otimes \mathrm{id}))$ for all bounded A.

Definition 5.2 (Von Neumann and entanglement entropy)**.** For a density operator with discrete spectral resolution $\rho = \sum_{\lambda>0} \lambda P_\lambda$, its *von Neumann entropy* is

$$S(\rho) = -k_B \operatorname{Tr}(\rho \log \rho) = -k_B \sum_{\lambda>0} (\operatorname{rank} P_\lambda) \lambda \log \lambda,$$

with $0 \log 0 = 0$; in infinite dimension the value may be $+\infty$. For a pure state $\rho = |\psi\rangle\langle\psi|$ on $H_A \otimes H_B$, its *entanglement entropy* across the bipartition is

$$S_A(\psi) = S(\operatorname{Tr}_B \rho).$$

The Schmidt spectral data will show that this also equals $S(\operatorname{Tr}_A \rho)$.

Exercise 5.1 (Classical and quantum Gibbs states)**.** Let K be n-dimensional, let $P_1, \ldots, P_n$ be mutually orthogonal rank-one projections with sum 1_K, and let $D = \operatorname{span}\{P_1, \ldots, P_n\} \cong \mathbb{C}^n$. Identify a state on D with a probability vector $p = (p_i)$ and the density operator $\rho_p = \sum_i p_i P_i$. Prove

$$S(\rho_p) = -k_B \sum_i p_i \log p_i,$$

so von Neumann entropy restricts to classical entropy on a commutative observable algebra. Prove more generally that $S(U\rho U^*) = S(\rho)$ for every unitary U.

Let $H = H^* \in B(K)$ have at least two distinct eigenvalues, and let the prescribed energy E lie strictly between its smallest and largest eigenvalues. Prove that there is a unique $\beta \in \mathbb{R}$ for which

$$\rho_\beta = \frac{e^{-\beta H}}{Z(\beta)}, \qquad Z(\beta) = \operatorname{Tr}(e^{-\beta H}), \qquad \operatorname{Tr}(\rho_\beta H) = E.$$

Prove strict concavity of S and use it with the normalization and energy constraints to show that ρ_β is their unique entropy maximizer. Derive

$$\operatorname{Tr}(\rho_\beta H) = -\partial_\beta \log Z, \qquad \operatorname{Var}_{\rho_\beta}(H) = \partial_\beta^2 \log Z.$$

Finally, restrict ρ_β to the commutative algebra generated by the spectral projections of H and recover the classical Gibbs probabilities, including their energy-level degeneracies. If g_i is the degeneracy of E_i and p_i is the total population of that energy level, derive

$$\frac{p_i}{p_j} = \frac{g_i}{g_j} e^{-\beta(E_i - E_j)}.$$

Explain why the degeneracy factor is absent for the ratio of populations of individual eigenstates, and compare both formulas with measured level-population ratios.

Exercise 5.2 (Schmidt projections and entanglement)**.** Let H_A and H_B be finite-dimensional. For a pure vector $\psi \in H_A \otimes H_B$, apply the projection form of the singular-value decomposition to the contraction map $C_\psi : H_A^* \to H_B$. If Q_s and R_s are its domain and range projections, decompose ψ into mutually orthogonal spectral blocks supported on $(Q_s H_A^*)^* \otimes R_s H_B$. Explain why a rank-one refinement within a block of multiplicity greater than one is not canonical. Prove that a pure state is separable exactly when its reduced state has rank one, and that the two reduced states have the same nonzero spectral values with multiplicities $\operatorname{rank} Q_s = \operatorname{rank} R_s$. Derive

$$S_A(\psi) = -k_B \sum_{s>0} (\operatorname{rank} Q_s) s^2 \log s^2$$

and prove that it vanishes exactly for separable pure states. Compute the joint and reduced states, local outcome frequencies, and entanglement entropy of a Bell state, and compare them with entangled-pair tomography.

Definition 5.3 (Representations and GNS construction)**.** A *representation* of a C^*-algebra A is a $*$-homomorphism $\pi : A \to B(H)$. For a state φ, define $N_\varphi = \{a : \varphi(a^*a) = 0\}$ and $\langle [a], [b] \rangle = \varphi(a^*b)$ on A/N_φ.

Exercise 5.3 (GNS theorem)**.** Complete A/N_φ and prove that left multiplication induces a cyclic representation $(\pi_\varphi, H_\varphi, \Omega_\varphi)$ satisfying $\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi \rangle$. Prove uniqueness up to a cyclic unitary equivalence.

Definition 5.4 (CAR algebra and fermionic Fock space)**.** For a complex Hilbert space K, the *canonical anticommutation relations algebra* $\mathrm{CAR}(K)$ is the unital C^*-algebra generated by conjugate-linear symbols $a(f)$ satisfying

$$\{a(f), a(g)\} = 0, \qquad \{a(f), a(g)^*\} = \langle f, g \rangle 1.$$

Let $\Lambda_2^n K$ be the Hilbert-space completion of the nth antisymmetric tensor power of K, with $\Lambda_2^0 K = \mathbb{C}$. The *antisymmetric Fock space* is the Hilbert direct sum

$$\mathcal{F}_-(K) = \bigoplus_{n \geq 0} \Lambda_2^n K.$$

Its vacuum is $\Omega = 1 \in \Lambda_2^0 K$; creation by f is exterior multiplication, and annihilation by f is its Hilbert-space adjoint.

Exercise 5.4 (CAR and GNS predict fermionic antibunching)**.** Prove that creation and annihilation on $\mathcal{F}_-(K)$ satisfy the CAR and that $\omega_0(x) = \langle \Omega, x\Omega \rangle$ is a state. Apply the preceding exercise to identify its GNS representation with the Fock representation. Let ω_C be a gauge-invariant quasifree state with one-particle density $0 \leq C \leq 1$, and use its determinantal Wick rule to prove, for orthonormal detector modes f, g, that

$$\omega_C(N_f N_g) = \omega_C(N_f)\omega_C(N_g) - |\langle f, Cg \rangle|^2, \qquad N_f = a(f)^* a(f).$$

Deduce $N_f^2 = N_f$ and fermionic antibunching. Compare the sign and separation dependence of the correlation deficit with the neutral-atom measurements in Fölling et al., *Nature* **444** (2006).

Part V

Observables

6 Operator Algebras, Channels, and Correlations

Experimental observations.

- An excited superconducting qubit relaxes approximately as e^{-t/T_1}; in the amplitude-damping limit its coherence decays as $e^{-t/(2T_1)}$.
- Loophole-free Bell experiments measure correlations that violate CHSH bounds satisfied by local hidden-variable models.

6.1 Operator Algebras and Local Observables

Definition 6.1 (Operator topologies and von Neumann algebras)**.** The *weak operator topology* is generated by $T \mapsto \langle x, Ty\rangle$; the *strong operator topology* is generated by $T \mapsto \|Tx\|$. For $S \subseteq B(H)$, its *commutant* is $S' = \{T : TS = ST \text{ for all } S \in S\}$. A *von Neumann algebra* is a unital $*$-subalgebra $M \subseteq B(H)$ with $M = M''$.

Exercise 6.1 (Bicommutant theorem)**.** Prove that for a unital $*$-subalgebra $A \subseteq B(H)$, the strong and weak operator closures of A equal A''. Deduce that von Neumann algebras have sufficient projections to approximate their self-adjoint elements spectrally.

Definition 6.2 (Normality and spatial tensor products)**.** A positive functional φ on a von Neumann algebra is *normal* if $a_\lambda \uparrow a$ implies $\varphi(a_\lambda) \uparrow \varphi(a)$ for every bounded increasing net of positive elements. A representation is *normal* if it preserves such suprema. For concrete von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, their *spatial tensor product* is

$$M \mathbin{\overline{\otimes}} N = (M \odot N)'' \subseteq B(H \otimes K).$$

Exercise 6.2 (Normal states and independent subsystems)**.** Show that every density operator ρ gives the normal state $a \mapsto \mathrm{Tr}(\rho a)$, and, using trace-class duality, prove the converse on $B(H)$. Prove $B(H) \mathbin{\overline{\otimes}} B(K) = B(H \otimes K)$ and identify product states and partial traces in this language. Explain why local normality—normality on every local algebra—is the infinite-system analogue of requiring ordinary reduced density matrices for all finite laboratories.

Definition 6.3 (Cyclic and separating vectors; modular theory)**.** For $M \subseteq B(H)$, a vector Ω is *cyclic* if $\overline{M\Omega} = H$ and *separating* if $a\Omega = 0$ implies $a = 0$. For such a pair (M, Ω), the antilinear operator $S_0(a\Omega) = a^*\Omega$ is closable. Write the polar decomposition of its closure as $S = J\Delta^{1/2}$. The positive operator Δ is the *modular operator*, and Tomita–Takesaki modular flow is

$$\sigma_t^\Omega(a) = \Delta^{it} a \Delta^{-it} \in M.$$

For a faithful density operator ρ on H, this flow on $B(H)$ is $\sigma_t^\rho(a) = \rho^{it} a \rho^{-it}$, and $K_\rho = -\log \rho$ is its *modular Hamiltonian.* A state φ is *β-KMS* for a flow α_t if, with the present sign convention and for analytic a, b,

$$\varphi(a\alpha_{t-i\beta}(b)) = \varphi(\alpha_t(b)a).$$

The vector state of Ω is 1-KMS for its modular flow.

Exercise 6.3 (Finite modular theory and entanglement tomography)**.** For finite-dimensional H, let $B(H)$ act on itself by left multiplication with inner product $\langle X, Y\rangle = \mathrm{Tr}(X^*Y)$, and set $\Omega_\rho = \rho^{1/2}$ for faithful ρ. Prove that Ω_ρ is cyclic and separating. For $S_0(a\Omega_\rho) = a^*\Omega_\rho$, compute the polar decomposition and show that its modular operator is $\Delta(X) = \rho X \rho^{-1}$, yielding the flow above. If ρ is a reduced state, recover its entanglement spectrum from K_ρ. Apply this reconstruction at successive times after a quench and compare the resulting spectrum with the trapped-ion data analyzed by Kokail et al., *Nature Physics* **17** (2021), and explain why finite-subsystem tomography does not by itself establish continuum modular theory.

Definition 6.4 (Traces and factor types)**.** A *finite trace* on a von Neumann algebra is a normal positive functional τ satisfying $\tau(ab) = \tau(ba)$. A possibly unbounded *normal tracial weight* is an additive, positively homogeneous map $\tau : M_+ \to [0, \infty]$ satisfying $\tau(x^*x) = \tau(xx^*)$ and $\tau(x_\lambda) \uparrow \tau(x)$ whenever $x_\lambda \uparrow x$. It is *semifinite* if every nonzero $x \in M_+$ majorizes a nonzero $y \in M_+$ with $\tau(y) < \infty$. Thus the standard traces on infinite type-I and type-II$_\infty$ factors are normal semifinite weights, not finite functionals.

Projections $p, q \in M$ are *Murray–von Neumann equivalent*, written $p \sim q$, if some $v \in M$ satisfies $v^*v = p$ and $vv^* = q$. A projection p is *finite* if $q \le p$ and $q \sim p$ imply $q = p$, and it is *minimal* if its only subprojections are 0 and p.

A *factor* is a von Neumann algebra with center $\mathbb{C}1$. A factor is *type I* if it has a nonzero minimal projection, equivalently if it is isomorphic to some $B(K)$; it is *type II* if it has nonzero finite projections but no minimal projections; and it is *type III* if it has no nonzero finite projection. Thus a type-III factor has no normal tracial state and no intrinsic trace-density representation of its normal states. A factor is *hyperfinite* if it is the weak closure of an increasing union of finite-dimensional $*$-subalgebras.

Exercise 6.4 (Projection dimension and factors)**.** For a finite factor with normalized trace τ, prove additivity and unitary invariance of τ on projections. Relate the possible projection values to matrix factors and to diffuse finite factors. Prove that $B(K)$ is type I and that a factor with a faithful normal tracial state cannot be type III. Explain why finite-dimensional quantum systems see only type-I factors, whereas the classification permits genuinely different infinite systems.

Definition 6.5 (Relative entropy of operator-algebraic states)**.** Let φ and ψ be faithful normal states on M, represented by vectors Ω_φ and Ω_ψ in standard form. Their relative Tomita operator is the closure of

$$S_{\varphi|\psi,0}(a\Omega_\psi) = a^*\Omega_\varphi,$$

and their *relative modular operator* is

$$\Delta_{\varphi|\psi} = S^*_{\varphi|\psi} S_{\varphi|\psi}.$$

The *Araki relative entropy* is the extended nonnegative number

$$D_M(\varphi\|\psi) = \langle \Omega_\varphi, \log \Delta_{\varphi|\psi}\, \Omega_\varphi\rangle.$$

For $M = B(H)$ and faithful density operators ρ, σ, the standard Hilbert–Schmidt representation gives $\Delta_{\rho|\sigma}(X) = \rho X \sigma^{-1}$ and hence

$$D(\rho\|\sigma) = \mathrm{Tr}(\rho(\log\rho - \log\sigma)).$$

Unlike the entropy of an individual density matrix, relative entropy and modular flow are intrinsic to an algebra and its states and remain meaningful for type-III factors.

Exercise 6.5 (Type-III localization and the split property)**.** Assume the structural theorem that, under the usual nontriviality, phase-space, and scaling hypotheses, local observable algebras in relativistic continuum quantum field theory are hyperfinite type III_1 factors, the full-modular-spectrum case of the type-III classification; see Yngvason, *Reports on Mathematical Physics* **55** (2005). Deduce that the restriction of the vacuum to a sharply bounded region has no intrinsic reduced density operator and that $-\operatorname{Tr}(\rho_A \log \rho_A)$ is therefore not an intrinsic continuum entropy. Explain how a split inclusion

$$\mathcal{A}(O_1) \subset F \subset \mathcal{A}(O_2), \qquad \overline{O_1} \subset O_2,$$

with F type I supplies a density-matrix regulator whose entropy depends on the collar $O_2 \setminus \overline{O_1}$ and on the choice of F. Contrast this regulator dependence and the usual ultraviolet divergence as the collar shrinks with the intrinsic relative entropy $D_{\mathcal{A}(O_1)}(\varphi \| \psi)$.

Exercise 6.6 (Bisognano–Wichmann flow and its experimental limit)**.** Let $W_R = \{x : x^1 > |x^0|\}$ be the right Rindler wedge of Minkowski space, and let Ω be the vacuum of a relativistic local net with stress-energy density T_{00}. Assume the Bisognano–Wichmann theorem

$$\Delta_{W_R}^{it} = U(\Lambda_{W_R}(-2\pi t)),$$

which identifies vacuum modular flow with the Lorentz boosts preserving W_R; see Bisognano–Wichmann, *Journal of Mathematical Physics* **16** (1975). In units $\hbar = c = k_B = 1$, derive on the time-zero half-space the modular Hamiltonian

$$K_{W_R} = 2\pi \int_{x^1>0} x^1 T_{00}(0, \mathbf{x})\, d^{d-1}x$$

up to an additive constant. Use its KMS property to obtain the Unruh temperature $T = a/(2\pi)$ for proper acceleration a.

Discretize the integral for a half-chain with local energy terms h_i and derive the lattice Bisognano–Wichmann ansatz in which the coefficient of h_i grows with its distance from the entanglement cut. Compare the predicted locality and spatial temperature profile with the entanglement Hamiltonians learned in the 51-ion XXZ-chain experiment of Joshi et al., *Nature* **624** (2023). Explain why this tests a finite-lattice consequence of modular QFT but cannot by itself establish that a continuum local algebra is type III. Identify the covariance, positive-energy, and vacuum hypotheses used by the modular-flow prediction that are not consequences of the factor type alone.

Definition 6.6 (Noncommutative probability)**.** A *noncommutative probability space* is a pair (M, φ) of a von Neumann algebra and a normal state; its random variables are operators affiliated with M.

Exercise 6.7 (Independent noncommutative subsystems)**.** For commuting von Neumann subalgebras $M_1, M_2 \subseteq M$, formulate independence in the state φ by factorization $\varphi(ab) = \varphi(a)\varphi(b)$. Compare this algebraic condition with a tensor-product state when $M_1 \vee M_2$ is spatially isomorphic to $M_1 \mathbin{\overline{\otimes}} M_2$, and explain why the split property is the relevant replacement for sharply adjacent local regions in continuum QFT.

6.2 Channels and Quantum Correlations

Channels describe how observables and states change under noise, measurement, and loss of access to an environment. Matrix amplification detects behavior that is invisible at the scalar level and ultimately controls nonlocal correlations.

Definition 6.7 (Positive and completely positive maps)**.** A linear map $\Phi : A \to B$ between C^*-algebras is *positive* if it preserves positive elements and *completely positive* if every $\mathrm{id}_{M_n} \otimes \Phi$ is positive. A *quantum channel* is a completely positive trace-preserving map on trace-class operators, equivalently a normal unital completely positive map on observables in the adjoint picture.

Exercise 6.8 (Stinespring and Kraus representations)**.** Prove that a unital completely positive map $\Phi : A \to B(H)$ has a representation $\Phi(a) = V^*\pi(a)V$. In finite dimensions, derive a Kraus representation $\Phi(a) = \sum_i K_i^* a K_i$ and its normalization. Prove that every channel is unitary evolution on a larger system followed by discarding an environment.

Exercise 6.9 (Amplitude damping reproduces spontaneous emission)**.** For a qubit with ground state $|0\rangle$, excited state $|1\rangle$, and environment vacuum $|0\rangle_E$, let

$$\begin{aligned} V_t|0\rangle &= |0\rangle|0\rangle_E, \\ V_t|1\rangle &= \sqrt{\eta_t}\,|1\rangle|0\rangle_E + \sqrt{1-\eta_t}\,|0\rangle|1\rangle_E, \qquad \eta_t = e^{-t/T_1}. \end{aligned}$$

Trace out the environment and derive

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta_t}|1\rangle\langle 1|, \qquad K_1 = \sqrt{1-\eta_t}|0\rangle\langle 1|.$$

Prove complete positivity, trace preservation, and the semigroup law. Derive $\rho_{11}(t) = e^{-t/T_1}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-t/(2T_1)}\rho_{01}(0)$. Compare the predicted decay and the environmental dependence of T_1 with the transmon measurements in Houck et al., *Physical Review Letters* **101** (2008).

Definition 6.8 (Measurements)**.** A *positive operator-valued measure* is a countably additive map E from measurable events to positive operators with $E(X) = 1$. In state ρ, its outcome probability is $\mathrm{Tr}(\rho E(B))$.

Exercise 6.10 (POVMs and dilation)**.** Prove that projection-valued measures are POVMs. Prove Naimark's dilation theorem in finite dimensions and derive the measurement channel associated to a POVM. Show that state discrimination can require a nonprojective POVM.

Definition 6.9 (Quantum codes and correctable errors)**.** Let $V : K \to H$ be an isometry between finite-dimensional Hilbert spaces. Its image $C = \operatorname{im} V$, with projection $P = VV^*$, is a *quantum code*; K is the logical system and H the physical system. Write $\mathcal{T}(H)$ for the trace-class operators on H. A channel $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ is *correctable on* C if there is a recovery channel $\mathcal{R}$ such that

$$(\mathcal{R} \circ \mathcal{N})(V\rho V^*) = V\rho V^*$$

for every state ρ on K. A *stabilizer code* is the common $+1$ spectral subspace of commuting self-adjoint unitaries. When $H = \bigotimes_{j=1}^n H_j$, the *distance* of C is the least number of tensor factors supporting an operator E for which PEP is not a scalar multiple of P, or $+\infty$ if there is no such operator.

Exercise 6.11 (Knill–Laflamme correction and decoupling)**.** Let $\mathcal{N}(\rho) = \sum_a E_a \rho E_a^*$ and let P project onto a code C. Prove that $\mathcal{N}$ is correctable on C exactly when there is a positive semidefinite matrix (λ_{ab}) such that

$$PE_a^*E_bP = \lambda_{ab}P \qquad \text{for every } a, b.$$

Show that this condition is independent of the chosen Kraus representation, and construct a recovery from the polar decomposition of the combined error map

$$W : C \otimes \ell^2(\{a\}) \longrightarrow H', \qquad W(\psi \otimes \delta_a) = E_a\psi,$$

together with the spectral projections of (λ_{ab}). Deduce that a code of distance d corrects every error supported on at most $\lfloor (d-1)/2 \rfloor$ tensor factors.

For an encoding $V : K \to H_A \otimes H_B$, prove that erasure of B is correctable exactly when the complementary channel $\rho \mapsto \mathrm{Tr}_A(V\rho V^*)$ is constant on logical states. Interpret this as the equivalence between recovery from A and absence of encoded information in B; see Knill–Laflamme, *Physical Review A* **55** (1997).

Definition 6.10 (Operator spaces and operator systems)**.** An *operator space* is a closed subspace $E \subseteq B(H)$ with matrix norms inherited from $M_n(E) \subseteq B(H^{\oplus n})$. A map $u : E \to F$ is *completely bounded* if $\|u\|_{cb} = \sup_n \| \mathrm{id}_{M_n} \otimes u\| < \infty$. An *operator system* is a closed self-adjoint linear subspace $S \subseteq B(H)$ containing the identity. Its matrix order is the family of cones $M_n(S)_+ = M_n(S) \cap B(H^{\oplus n})_+$.

Definition 6.11 (The confusability operator system of a channel)**.** Let H, H' be finite-dimensional and let $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ be a channel with Kraus operators $E_a : H \to H'$. Its *confusability operator system* is

$$S_{\mathcal{N}} = \mathrm{span}\{E_a^* E_b : a, b\} \subseteq B(H).$$

It is self-adjoint, and it contains the identity because $\sum_a E_a^* E_a = 1_H$.

Exercise 6.12 (Kraus independence of the confusability system)**.** If $\{E_a\}$ and $\{F_\mu\}$ are two Kraus representations of $\mathcal{N}$, pad the smaller family by zero operators and prove that there is a unitary matrix $(u_{\mu a})$ such that $F_\mu = \sum_a u_{\mu a} E_a$. Use this relation in both directions to prove

$$\mathrm{span}\{F_\mu^* F_\nu\} = \mathrm{span}\{E_a^* E_b\}.$$

Thus $S_{\mathcal{N}}$ depends only on the channel, not on a choice of environment coordinates. Interpret it as the noncommutative analogue of a classical channel's confusability graph; see Weaver, *Quantum Graphs as Quantum Relations*, Journal of Geometric Analysis **31** (2021).

Definition 6.12 (Protected observable algebras)**.** Let P project onto a code $C \subseteq H$. A unital $*$-subalgebra $\mathcal{A} \subseteq PB(H)P$, whose identity is P, is *protected* or *correctable* for $\mathcal{N}$ if there is a recovery channel $\mathcal{R} : \mathcal{T}(H') \to \mathcal{T}(H)$ such that

$$P(\mathcal{R} \circ \mathcal{N})^*(A)P = A \qquad (A \in \mathcal{A}).$$

Thus every observable in $\mathcal{A}$, rather than necessarily every encoded state, can be recovered. Write $\mathcal{A}' = \{X \in PB(H)P : [X, A] = 0 \text{ for all } A \in \mathcal{A}\}$ for the commutant in the code corner.

Exercise 6.13 (Exact operator-algebra quantum error correction)**.** Prove that $\mathcal{A}$ is correctable for $\mathcal{N}$ if and only if

$$[PE_a^* E_b P, A] = 0 \qquad (a, b,\ A \in \mathcal{A}),$$

or, equivalently,

$$PS_{\mathcal{N}}P \subseteq \mathcal{A}'.$$

Derive necessity by applying a Stinespring dilation to the recovery identity, and prove sufficiency by decomposing the finite-dimensional algebra and constructing a recovery on each summand. For $\mathcal{A} = PB(H)P \cong B(C)$, use $\mathcal{A}' = \mathbb{C}P$ to recover the ordinary Knill–Laflamme equations $PE_a^* E_b P = \lambda_{ab} P$. Show that a commutative algebra protects a classical label, while a decomposition

$$\mathcal{A} \cong \bigoplus_\alpha (B(H_\alpha^A) \otimes 1_{H_\alpha^B})$$

protects hybrid information: the central label α is classical, the H_α^A factor is quantum, and H_α^B is an unprotected gauge subsystem. Compare with Bény–Kempf–Kribs, *Quantum Error Correction of Observables*, Physical Review A **76** (2007).

Exercise 6.14 (Normal channels and von Neumann protected algebras)**.** Let H, H' be arbitrary Hilbert spaces, let $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ be a channel whose adjoint is normal, and choose a Stinespring isometry $V : H \to H' \otimes K$. Define the complementary channel and its adjoint by

$$\mathcal{N}^c(\rho) = \mathrm{Tr}_{H'}(V\rho V^*), \qquad (\mathcal{N}^c)^*(Y) = V^*(1_{H'} \otimes Y)V.$$

For a projection P and a von Neumann algebra $\mathcal{A} \subseteq PB(H)P$, define correctability using a recovery channel with normal adjoint and the same observable identity as above. Prove the infinite-dimensional exact criterion

$$P(\mathcal{N}^c)^*(B(K))P \subseteq \mathcal{A}'.$$

Show from uniqueness of minimal Stinespring dilations that this condition is independent of the chosen complementary channel. Explain how it replaces a Kraus-index calculation when the environment is not finite-dimensional and allows protected algebras of every von Neumann type; see Bény–Kempf–Kribs, *Quantum Error Correction on Infinite-Dimensional Hilbert Spaces*, Journal of Mathematical Physics **50** (2009).

Definition 6.13 (Ancilla-assisted channel discrimination)**.** A *one-use discrimination strategy* for channels $\Phi_1, \ldots, \Phi_m$ prepares a state on $H \otimes K$, applies $\Phi_j \otimes \mathrm{id}_K$ to the unknown channel input, and measures the output with a POVM. The strategy is *ancilla assisted* when $\dim K > 1$. It is *unambiguous* when every conclusive outcome identifies the channel without error. Its *inconclusive probability* is denoted P_I.

Exercise 6.15 (An ancilla improves measurement-channel discrimination)**.** For $0 < \theta < \pi/4$, define

$$|\phi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \quad |\psi\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle,$$

and let $\mathcal{M}, \mathcal{N}$ be the two projective measurement channels with rank-one projection families $\{P_\phi, P_{\phi^\perp}\}$ and $\{P_\psi, P_{\psi^\perp}\}$. Use a singlet probe, retain the second qubit as an ancilla, and condition on the classical measurement outcome. Reduce the task to unambiguous discrimination of $|\phi\rangle$ and $|\psi\rangle$ and prove

$$P_I^{\mathrm{anc}} = \cos(2\theta), \qquad P_I^{\mathrm{sep}} = \frac{1 + \cos^2(2\theta)}{2}$$

for the optimal ancilla-assisted and single-qubit strategies. Deduce the strict advantage for $0 < \theta < \pi/4$ and express it as a distinction between the first and amplified matrix levels of the measurement channels. Compare the predicted success and inconclusive probabilities with Miková et al., *Physical Review A* **90** (2014).

Exercise 6.16 (Concrete matrix-norm axioms)**.** Prove that concrete operator-space norms satisfy

$$\|x \oplus y\| = \max(\|x\|, \|y\|), \qquad \|axb\| \le \|a\|\|x\|\|b\|.$$

Track the matrix sizes and prove that completely bounded maps preserve both direct sums and multiplication by scalar matrices at every level.

Exercise 6.17 (Hahn–Banach separation)**.** Let Y be a linear subspace of a complex normed space X. Use Zorn's lemma, first over the underlying real spaces and then after complexification, to prove that every bounded linear functional on Y extends to X without increasing its norm. Deduce that each nonzero $x \in X$ has a norm-one functional $f \in X^*$ with $f(x) = \|x\|$.

Exercise 6.18 (Matrix Hahn–Banach separation)**.** Let E carry matrix norms satisfying the two concrete matrix-norm axioms, and let $F \in M_n(E)^*$ have norm at most one. Apply geometric Hahn–Banach separation to the matrix unit ball and prove that there are states p, q on M_n such that

$$|F(ayb)| \le p(aa^*)^{1/2} \, \|y\|_n \, q(b^*b)^{1/2} \qquad (a, b \in M_n,\ y \in M_n(E)).$$

Represent p and q by unit vectors in amplifications of $\mathbb{C}^n$. Use the displayed inequality to construct a bounded sesquilinear form on their cyclic subspaces and hence a contraction $T : M_n(E) \to M_n(M_n)$ satisfying

$$T(ayb) = (a \otimes 1)T(y)(b \otimes 1), \qquad F(y) = \langle T(y)\xi, \eta \rangle.$$

Prove from the bimodule identity that $T = \phi^{(n)}$ for a linear map $\phi : E \to M_n$, and use the matrix-norm axioms to show that $\|\phi\|_{cb} = \|\phi^{(n)}\| \le 1$. Finally, for each $x \in M_n(E)$ choose a norming F and deduce

$$\|\phi^{(n)}(x)\| = \|x\|_n.$$

Exercise 6.19 (Ruan's representation theorem)**.** Prove conversely that every vector space with matrix norms satisfying these axioms admits a complete isometry into $B(H)$. Apply matrix Hahn–Banach separation to a norming family of matrices and take the direct sum of the resulting maps.

Exercise 6.20 (Finite-dimensional Arveson–Wittstock extension)**.** Prove in finite dimensions that a completely positive map from an operator system extends completely positively to the containing C^*-algebra. Deduce, by the 2×2 operator-system construction, that completely bounded maps between finite-dimensional operator spaces admit norm-preserving extensions.

Definition 6.14 (Operator-space tensor products)**.** The *minimal operator-space tensor norm* on $E \otimes F$ is inherited from concrete inclusions in $B(H \otimes K)$. The *Haagerup norm* is

$$\|z\|_h = \inf\{\|[x_1 \ \cdots \ x_r]\| \, \|[y_1 \ \cdots \ y_r]^T\| : z = \sum_i x_i \otimes y_i\}.$$

A bilinear form $u : E \times F \to \mathbb{C}$ is *jointly completely bounded* when the maps

$$u_n([x_{ij}], [y_{kl}]) = [u(x_{ij}, y_{kl})]_{(i,k),(j,l)}$$

from $M_n(E) \times M_n(F)$ to M_{n^2} are uniformly bounded; the supremum of their norms is $\|u\|_{jcb}$.

Exercise 6.21 (Tensor norms linearize bilinear maps)**.** Prove the universal property of the projective Banach-space tensor norm. Prove that the Haagerup tensor product linearizes completely bounded factorizations and compare the minimal, Haagerup, and projective norms on finite-dimensional matrix spaces.

Exercise 6.22 (Noncommutative Grothendieck inequality)**.** Assume that for every bounded bilinear form $u : A \times B \to \mathbb{C}$ on unital C^*-algebras there are states f_1, f_2 on A and g_1, g_2 on B such that

$$|u(a, b)| \le \|u\| \big(f_1(a^*a) + f_2(aa^*)\big)^{1/2} \big(g_1(b^*b) + g_2(bb^*)\big)^{1/2}.$$

Specialize this inequality to commutative C^*-algebras and recover the classical Grothendieck domination of bounded bilinear forms. Apply it to matrix algebras and express the two quadratic terms as row and column norms. Use Haagerup, *The Grothendieck inequality for bilinear forms on C^*-algebras.*

Exercise 6.23 (Operator-space Grothendieck inequality)**.** Assume that for every jointly completely bounded bilinear form $u : A \times B \to \mathbb{C}$ on unital C^*-algebras there are states f_1, f_2 on A and g_1, g_2 on B such that

$$|u(a,b)| \leq \|u\|_{jcb} \left(f_1(aa^*)^{1/2} g_1(b^*b)^{1/2} + f_2(a^*a)^{1/2} g_2(bb^*)^{1/2} \right),$$

as proved by Haagerup–Musat, *The Effros–Ruan conjecture for bilinear forms on C^*-algebras*. Use matrix amplification to explain why $\|u\|_{jcb}$, rather than only $\|u\|$, controls coupling to arbitrary ancillas. Give examples showing that both row and column terms are necessary.

Definition 6.15 (Bell functionals and nonlocal games)**.** A *Bell functional* is an array $M = (M_{xy}^{ab})$ evaluated on conditional distributions by $\langle M, p \rangle = \sum_{a,b,x,y} M_{xy}^{ab} p(a, b \mid x, y)$. A two-player *nonlocal game* $G = (\pi, V)$ consists of a question distribution $\pi(x, y)$ and a payoff $0 \leq V(a, b \mid x, y) \leq 1$. Its classical and entangled values $\omega(G)$ and $\omega^*(G)$ are the optimal expected payoffs using, respectively, shared randomness and local POVMs $\{E_a^x\}_a, \{F_b^y\}_b$ on a shared finite-dimensional state ρ, with $p(a, b \mid x, y) = \mathrm{Tr}(\rho(E_a^x \otimes F_b^y))$. An *XOR game* has binary answers and scores only their parity, equivalently a correlation $\sum_{x,y} M_{xy} \mathbb{E}[a_x b_y]$ for signs a_x, b_y.

Exercise 6.24 (Magic-square pseudotelepathy)**.** The referee chooses $(x, y) \in \{1, 2, 3\}^2$ uniformly. Alice returns a row $a \in \{\pm 1\}^3$ with product $+1$, Bob returns a column $b \in \{\pm 1\}^3$ with product -1, and they win when $a_y = b_x$. Prove by multiplying all row and column constraints that no deterministic classical strategy wins every question, construct one losing on only one question, and hence show $\omega(G_{\mathrm{MS}}) = 8/9$. Using two Bell pairs and a Mermin–Peres square of commuting two-qubit Pauli observables, construct a strategy with $\omega^*(G_{\mathrm{MS}}) = 1$.

Compare the ideal query-by-query predictions with the hyperentangled-photon experiment of Xu et al., *Physical Review Letters* **129** (2022). Explain separately the locality and fair-sampling assumptions in that experiment. Show why a score above 8/9 is a device-independent witness under the stated assumptions; then, assuming a robust magic-square rigidity theorem, explain how a score $1 - \varepsilon$ certifies two Bell pairs up to local isometries and errors controlled by ε.

Definition 6.16 (The tensor of a game)**.** Set

$$E_{X,A} = \ell_1^X(\ell_\infty^A), \qquad \|z\|_{E_{X,A}} = \sum_x \max_a |z_{x,a}|,$$

with its standard dual operator-space structure. The tensor associated to $G = (\pi, V)$ is

$$\widehat{G} = \sum_{x,y,a,b} \pi(x,y) V(a, b \mid x, y)(e_x \otimes e_a) \otimes (e_y \otimes e_b) \in E_{X,A} \otimes E_{Y,B}.$$

Exercise 6.25 (Game values as Banach and operator-space norms)**.** Identify deterministic strategies with extreme points of the relevant unit balls and prove

$$\omega(G) = \|\widehat{G}\|_{E_{X,A} \otimes_\varepsilon E_{Y,B}}, \qquad \omega^*(G) = \|\widehat{G}\|_{E_{X,A} \otimes_{\min} E_{Y,B}}.$$

Equivalently, show that these are the bounded and completely bounded norms of the bilinear payoff form on the dual mixed-norm spaces. Locate in the proof the use of positivity of a game tensor, and explain why the same simple formula need not hold for an arbitrary signed Bell functional. Use Palazuelos–Vidick, *Survey on Nonlocal Games and Operator Space Theory*, *Journal of Mathematical Physics* **57** (2016) as a guide.

Exercise 6.26 (Grothendieck bounds for XOR games)**.** For an XOR game, prove that its classical and entangled biases are

$$\beta(G) = \sup_{a_x, b_y = \pm 1} \left| \sum M_{xy} a_x b_y \right|, \quad \beta^*(G) = \sup_{\|u_x\|, \|v_y\| \le 1} \left| \sum M_{xy} \langle u_x, v_y \rangle \right|.$$

Use Tsirelson's vector representation and the real Grothendieck inequality to prove $\beta^*(G) \le K_G^{\mathbb{R}} \beta(G)$. For CHSH, compute the classical bound 2 and quantum bound $2\sqrt{2}$.

Exercise 6.27 (Unbounded advantages for multi-output games)**.** For general multi-output games, take as supplied the Khot–Vishnoi estimates $\omega(G_{\mathrm{KV}}) \le C/n$ and $\omega^*(G_{\mathrm{KV}}) \ge c/(\log n)^2$, achieved with local entanglement dimension n. Use the preceding norm dictionary to deduce

$$\frac{\|\widehat{G}_{\mathrm{KV}}\|_{\min}}{\|\widehat{G}_{\mathrm{KV}}\|_{\varepsilon}} \ge c' \frac{n}{(\log n)^2}.$$

Explain why the XOR comparison remains dimension independent while general game tensors require operator-space matrix levels. Consult Buhrman–Regev–Scarpa–de Wolf, *Theory of Computing* **8** (2012) for the supplied estimates.

Exercise 6.28 (Measured nonclassical correlations)**.** For a polarization-entangled two-photon state, choose four polarization angles attaining the CHSH value $2\sqrt{2}$ and derive the predicted coincidence correlations. As experimental input, use the adjusted local-realistic p-value 2.3×10^{-7} reported by Shalm et al., *Physical Review Letters* **115** (2015). Determine at the 5%, 1%, and five-standard-deviation significance levels, using the one-sided Gaussian five-sigma threshold 2.87×10^{-7}, whether the local-realistic null is excluded. State the tested freedom-of-choice and spacelike-separation assumptions and explain why the result does not exclude arbitrary superdeterministic or signaling models.

Part VI

Topology

7 Topology, K-Theory, and Index

Experimental observations.

- In the integer quantum Hall regime, transverse conductance forms plateaus at integer multiples of e^2/h while longitudinal resistance is strongly suppressed.

7.1 Homotopy, Cohomology, and Bundles

Definition 7.1 (Homotopy and fundamental group)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $H : X \times [0, 1] \to Y$ with endpoints f and g. The *fundamental group* $\pi_1(X, x_0)$ is the group of endpoint-fixed homotopy classes of loops based at x_0, with concatenation.

Exercise 7.1 (Fundamental groups and winding)**.** Prove invariance of π_1 under pointed homotopy equivalence and show that an unpointed homotopy changes the induced map only by the appropriate change-of-basepoint isomorphism. Prove the Seifert–van Kampen theorem for a union with path-connected intersection. Compute $\pi_1(S^1) \cong \mathbb{Z}$ by lifting loops to $\mathbb{R}$, and prove that the resulting integer is the winding number.

Definition 7.2 (Homology and cohomology)**.** Fix a commutative ring R with identity. The *singular chain group* $C_n(X;R)$ is the free R-module on continuous maps $\Delta^n \to X$, with alternating face boundary ∂. The *homology* is $H_n(X;R) = \ker \partial_n / \operatorname{im} \partial_{n+1}$. The *cochain complex* is $C^n(X;R) = \operatorname{Hom}_R(C_n(X;R), R)$ with coboundary $\delta\phi = \phi\partial$, and its cohomology is $H^n(X;R)$.

Definition 7.3 (Cofibers, finite CW complexes, and cohomology theories)**.** A based inclusion $A \hookrightarrow X$ is a *cofibration* if every homotopy on A that extends at its initial time to X extends over the whole homotopy. Its *cofiber* is X/A. More generally, the cofiber of a based map $f : A \to X$ is the pushout $X \cup_f CA$, where CA is the reduced cone; the reduced suspension is $\Sigma A = CA/A$.

A *finite CW complex* is obtained from the empty space by finitely many pushouts

$$\begin{array}{ccc} \coprod_\alpha S^{n-1} & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_\alpha D^n & \longrightarrow & X^n, \end{array}$$

where X^n is the nth skeleton. A *reduced cohomology theory* on based finite CW complexes is a graded contravariant functor $\widetilde{E}^*$ that is homotopy invariant, vanishes on a point, takes finite wedges to direct sums, has natural suspension isomorphisms $\widetilde{E}^{q+1}(\Sigma X) \cong \widetilde{E}^q(X)$, and makes

$$\widetilde{E}^q(X/A) \longrightarrow \widetilde{E}^q(X) \longrightarrow \widetilde{E}^q(A)$$

exact for every cofibration $A \hookrightarrow X$.

Definition 7.4 (Products and pairings)**.** The *Kronecker pairing* is $H^n(X;R) \times H_n(X;R) \to R$, $([\phi],[c]) \mapsto \phi(c)$. The *cup product* is the product on cohomology induced by the front–back face formula on cochains.

Exercise 7.2 (Homology, cohomology, and degree)**.** Prove $\partial^2 = 0$, $\delta^2 = 0$, and homotopy invariance. For a CW pair (X, A), define $C_*(X, A; R) = C_*(X;R)/C_*(A;R)$, construct barycentric subdivision, and use it to prove excision and $H^q(X, A; R) \cong \widetilde{H}^q(X/A; R)$. Construct the Puppe sequence of a cofibration and derive the long exact sequence of a pair and the Mayer–Vietoris sequence. Deduce that reduced singular cohomology is a reduced cohomology theory.

Prove that $X^{n-1} \hookrightarrow X^n$ is a cofibration and X^n/X^{n-1} is a finite wedge of n-spheres. By induction over the skeleta, prove that a natural transformation between reduced cohomology theories that is an isomorphism on S^0 in every degree is an isomorphism on every finite CW complex; prove the required five-term diagram chase. Compute the homology and cohomology rings of spheres and tori. For a continuous map $f : S^n \to S^n$, define its degree by $f_*[S^n] = \deg(f)[S^n]$ and prove multiplicativity and agreement with winding number for $n = 1$.

Exercise 7.3 (Topological bundles from transition functions)**.** Forget the smooth structure on the bundles introduced with gauge fields and retain their continuous transition maps. Prove that transition functions satisfying the cocycle condition construct a topological bundle and that changes of local trivialization give isomorphic bundles. Prove that a principal bundle is trivial exactly when it has a global section. Construct associated vector bundles from representations of G.

Definition 7.5 (Global connections and holonomy)**.** A *connection* on a vector bundle $E \to M$ is a map $\nabla : \Gamma(E) \to \Omega^1(M; E)$ satisfying $\nabla(fs) = df \otimes s + f\nabla s$. Its curvature is $F_\nabla = \nabla^2 \in \Omega^2(M; \operatorname{End} E)$. Its *holonomy* along a loop is the automorphism of the initial fiber given by parallel transport.

Exercise 7.4 (Curvature and holonomy)**.** Derive local connection and curvature forms and their gauge-transformation laws. Prove the Bianchi identity. Show that infinitesimal holonomy is curvature and compute the $U(1)$ holonomy around the boundary of a surface as $\exp(i \int F)$ when a global potential is available.

Exercise 7.5 (Aharonov–Bohm holonomy)**.** On $X = \mathbb{R}^2 \setminus \{0\}$, consider the $U(1)$ gauge potential $A = (\Phi/2\pi)d\theta$. Verify that $F = dA = 0$ on X but $\oint_\gamma A = \Phi$ for a loop of winding number one. Show that a particle of charge q acquires the gauge-invariant phase $\exp(iq\Phi/\hbar)$, derive the relative phase of two paths enclosing the excluded flux, and obtain the flux period $h/|q|$. Prove that A cannot be removed by a single-valued gauge transformation unless its holonomy is trivial. Compare the predicted fringe displacement with the shielded-flux electron-holography experiment of Tonomura et al., *Physical Review Letters* **56** (1986). Explain why the result establishes the physical significance of holonomy, not of a gauge-dependent potential by itself, even though curvature vanishes along the electron paths.

Definition 7.6 (Grassmannians and universal bundles)**.** For a finite-dimensional Hermitian space V, the *Grassmannian* $\mathrm{Gr}_n(V)$ is the space of n-dimensional subspaces of V. Its *tautological bundle* has fiber W over $W \in \mathrm{Gr}_n(V)$. Isometric inclusions of Hermitian spaces induce compatible inclusions of their Grassmannians and tautological bundles, and

$$BU(n) = \varinjlim_N \mathrm{Gr}_n(\mathbb{C}^N), \qquad \gamma_n = \varinjlim_N \gamma_{n,N}.$$

Exercise 7.6 (Classifying maps)**.** Let $E \to X$ be a rank-n complex vector bundle over a compact Hausdorff space. Construct a finite trivializing cover, a subordinate partition of unity, a Hermitian metric, and a fiberwise isometric embedding $E \hookrightarrow X \times \mathbb{C}^N$. Show that its ranges define a continuous map $f_E : X \to \mathrm{Gr}_n(\mathbb{C}^N)$ with $E \cong f_E^* \gamma_{n,N}$. Prove that two choices of embedding become homotopic after stabilization by placing them in orthogonal summands, and prove that homotopic maps have isomorphic pullbacks. Deduce the bijection

$$[X, BU(n)] \longleftrightarrow \{\text{rank-}n \text{ complex bundles over } X\}/\cong .$$

Exercise 7.7 (Universal Chern classes)**.** Identify $BU(1)$ with $\mathbb{P}(\mathbb{C}^\infty)$ and use its even-dimensional cell decomposition to compute $H^*(BU(1); \mathbb{Z}) = \mathbb{Z}[x]$, where x evaluates as -1 on the projective line carrying the tautological line bundle. Let

$$s : BU(1)^n \longrightarrow BU(n)$$

classify the direct sum of the n tautological lines, and set $x_i = \mathrm{pr}_i^* x$. Construct the Schubert cell decompositions of the finite Grassmannians and their flag bundles. Use the cellular filtration and the splitting map s to prove that s^* is injective with image

$$\mathbb{Z}[x_1, \ldots, x_n]^{\mathfrak{S}_n} = \mathbb{Z}[e_1(x), \ldots, e_n(x)].$$

Define $c_j(\gamma_n) \in H^{2j}(BU(n); \mathbb{Z})$ by $s^* c_j(\gamma_n) = e_j(x_1, \ldots, x_n)$. For the block-sum map $BU(n) \times BU(m) \to BU(n+m)$, prove

$$c(\gamma_n \oplus \gamma_m) = c(\gamma_n)c(\gamma_m), \qquad c = 1 + c_1 + c_2 + \cdots .$$

Definition 7.7 (Chern classes and character)**.** If a rank-n complex vector bundle $E \to X$ is classified by $f : X \to BU(n)$, its *Chern classes* are $c_j(E) = f^* c_j(\gamma_n) \in H^{2j}(X; \mathbb{Z})$. If the formal roots of $1 + c_1(E) + \cdots + c_n(E)$ are $y_1, \ldots, y_n$, the *Chern character* is the degreewise class

$$\mathrm{ch}(E) = \sum_{i=1}^n e^{y_i} \in \prod_{k \geq 0} H^{2k}(X; \mathbb{Q}),$$

where each symmetric polynomial in the y_i is expressed in the Chern classes by the Newton identities and the definition extends additively to $K^0(X)$.

Exercise 7.8 (Naturality, sums, and the Chern character)**.** Prove that the classifying-map definition is independent of every choice and that $c_j(g^*E) = g^*c_j(E)$. Use the block-sum calculation above to prove the Whitney formula $c(E \oplus F) = c(E)c(F)$. Prove that the Chern character is a natural ring map and that $\mathrm{ch}(L) = e^{c_1(L)}$ for a line bundle.

Exercise 7.9 (Curvature representatives)**.** For a Hermitian bundle with unitary connection, prove that

$$\det\left(1 + \frac{i}{2\pi}F_\nabla\right) \quad \text{and} \quad \operatorname{Tr}\exp\left(\frac{i}{2\pi}F_\nabla\right)$$

are closed forms whose cohomology classes are independent of the connection. Identify their de Rham classes with the images of the Chern classes and Chern character, and prove their Whitney-sum formulas.

Definition 7.8 (Berry geometry)**.** For a smooth family of one-dimensional spectral projections P_x on a Hilbert space, the ranges form a line bundle. The *Berry connection* is $\nabla = P \circ d$ and its curvature is the *Berry curvature*. Its holonomy is the *Berry phase*. The integral $\langle c_1(L), [\Sigma]\rangle$ is the *Chern number* over an oriented closed surface Σ.

Exercise 7.10 (Berry phase and quantized response)**.** Compute the Berry curvature of a two-level Hamiltonian $H(n) = n \cdot \sigma$ over S^2 for both the lower and upper eigenline bundles. With the standard orientation on S^2, compute their opposite Chern numbers and state the sign convention used. Identify the two eigenlines with classifying maps $f_\pm : S^2 \to \mathbb{P}(\mathbb{C}^2)$ and, for the normalization in Exercise 4.12, prove

$$F^{\pm}_{\mathrm{Berry}} = -2i f_\pm^* \omega_{\mathrm{FS}}.$$

For a gapped two-dimensional periodic Hamiltonian, derive the Kubo formula expressing Hall conductance as e^2/h times the occupied-band Chern number. Deduce integer quantization under gap-preserving perturbations and compare with integer quantum Hall plateaus. For an adiabatic closed parameter loop, compute the Berry holonomy, show that it is independent of traversal rate, and identify the corresponding interference-fringe shift.

7.2 K-Theory and Index

Cohomology assigns additive classes to bundles. K-theory first makes bundle addition invertible and then connects stable topology to the analytic index of differential operators.

Definition 7.9 (Grothendieck completion and stable equivalence)**.** The *Grothendieck group* of a commutative monoid M is the universal abelian group receiving a monoid map from M. Vector bundles E, F are *stably equivalent* if $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ for some m, n; this relation forgets rank and therefore describes reduced stable classes. The group $K^0(X)$ is the Grothendieck group of complex vector bundles on compact X. For based X, let $\widetilde{K}^0(X)$ be the kernel of restriction to the basepoint and, for every $j \geq 0$, define

$$\widetilde{K}^j(X) = \widetilde{K}^0(\Sigma^j X).$$

For unbased X, write $X_+ = X \sqcup \{*\}$ and set $K^j(X) = \widetilde{K}^j(X_+)$ for $j \geq 0$.

Definition 7.10 (Relative and compactly supported K-theory)**.** For a compact pair (X, A) and a locally compact space Y, define

$$K^j(X,A) = \widetilde{K}^j(X/A), \qquad K^j_c(Y) = \widetilde{K}^j(Y^+),$$

where Y^+ is the one-point compactification. A triple (F^0, F^1, σ) of vector bundles on Y, with $\sigma : F^0 \to F^1$ invertible off a compact set, represents a class in $K^0_c(Y)$ up to homotopy and stabilization.

Exercise 7.11 (Exactness of K-theory)**.** Prove homotopy invariance and finite-wedge additivity for reduced K-theory. For a cofibration $A \hookrightarrow X$, use clutching of bundles to prove exactness of the cofiber sequence in all nonnegative degrees defined above. Derive the relative and Mayer–Vietoris sequences from the general cofiber construction, and prove that external product is compatible with them.

Definition 7.11 (The K-theory Thom class)**.** Let $\pi : E \to X$ be a rank-r Hermitian complex vector bundle. Exterior multiplication by v and its adjoint give the odd endomorphism

$$c(v) = \varepsilon(v) - \iota(v) : \Lambda^* E_{\pi(v)} \longrightarrow \Lambda^* E_{\pi(v)}, \qquad c(v)^2 = -\|v\|^2.$$

Writing $c^+(v)$ for its even-to-odd part, the *K-theory Thom class* is

$$u_E = [\pi^*\Lambda^{\mathrm{even}}E, \pi^*\Lambda^{\mathrm{odd}}E, c^+] \in K^0_c(E).$$

Exercise 7.12 (Clutching and the Bott class)**.** Prove the clutching description of complex vector bundles on suspensions and spheres. For the complex line over a point, identify $u_{\mathbb{C}}$ under $\mathbb{C}^+ \cong S^2$ with

$$\beta = [H] - [\underline{\mathbb{C}}] \in \widetilde{K}^0(S^2),$$

where H is the Hopf line. Prove that β generates $\widetilde{K}^0(S^2)$ and that $\widetilde{K}^0(S^3) = 0$. Deduce that the Thom map for a point is an isomorphism in degrees zero and one.

Exercise 7.13 (Thom isomorphism and Bott periodicity)**.** Apply K-theory exactness and cellular induction to the preceding point calculation and prove the trivial-line result over finite CW complexes. Iterate it for trivial bundles and then use Mayer–Vietoris over bundle trivializations to prove the complex Thom isomorphism

$$K^j(X) \longrightarrow K^j_c(E), \qquad a \longmapsto \pi^* a\, u_E, \qquad j = 0, 1,$$

for finite CW complexes X. For the trivial line over a compact based finite CW complex, identify its relative Thom space with $S^2 \wedge X$ and deduce complex Bott periodicity:

$$\widetilde{K}^j(X) \xrightarrow{\boxtimes\beta} \widetilde{K}^j(S^2 \wedge X)$$

is an isomorphism.

Exercise 7.14 (K-theory of spheres)**.** Use Bott periodicity and the clutching calculation to compute the complex K-groups of spheres. Extend K^j to negative degrees by two-periodicity and use K-theory exactness to prove that it is a reduced cohomology theory on finite CW complexes.

Exercise 7.15 (K-theory of tori and class-A phases)**.** Derive from the cofibration for $X \times S^1$ and Bott periodicity the split formula $K^j(X \times S^1) \cong K^j(X) \oplus K^{j-1}(X)$, with indices taken modulo two, and compute the complex K-groups of tori. Show that a gapped family of finite-rank Hamiltonians defines the stable class of its occupied bundle and identify the strong and weak class-A invariants on Brillouin tori in dimensions one through three.

Definition 7.12 (Operator K-theory)**.** For a unital C^*-algebra A, $K_0(A)$ is the Grothendieck group of stable homotopy classes of projections in matrix algebras over A, where orthogonal sum gives the underlying monoid, and $K_1(A)$ is the stable homotopy group of unitaries in matrix algebras over A.

Exercise 7.16 (Topological and operator K-theory)**.** Construct the Serre–Swan correspondence between vector bundles over compact X and finitely generated projective $C(X)$-modules. Deduce $K^j(X) \cong K_j(C(X))$ and formulate Bott periodicity for C^*-algebras.

Definition 7.13 (Compact and Fredholm operators)**.** An operator on a Hilbert space is *compact* if it is a norm limit of finite-rank operators. A bounded operator $T : H_0 \to H_1$ is *Fredholm* if its kernel and cokernel are finite-dimensional and its range is closed; its *index* is $\operatorname{ind} T = \dim\ker T - \dim\operatorname{coker} T$. The *Calkin algebra* is $\mathcal{Q}(H) = B(H)/\mathcal{K}(H)$.

Exercise 7.17 (Fredholm parametrices and the Calkin algebra)**.** For $T : H_0 \to H_1$, prove that T is Fredholm exactly when there is a bounded $S : H_1 \to H_0$ such that $ST - \mathrm{id}_{H_0}$ and $TS - \mathrm{id}_{H_1}$ are compact. When $H_0 = H_1 = H$, identify this condition with invertibility of the image of T in $\mathcal{Q}(H)$. Prove homotopy invariance, compact-perturbation invariance, and additivity of the index.

Exercise 7.18 (Shift index and the boundary map)**.** On an infinite-dimensional separable Hilbert space, compute the index of the unilateral shift and identify the connecting map $K_1(\mathcal{Q}(H)) \to K_0(\mathcal{K}(H))$ from the Calkin extension with the Fredholm index.

Definition 7.14 (Elliptic symbols)**.** For vector bundles E, F over a compact smooth manifold, the *principal symbol* of an order-m differential operator $D : \Gamma(E) \to \Gamma(F)$ is the homogeneous bundle map $\sigma_D : \pi^*E \to \pi^*F$ on T^*M. The operator is *elliptic* if $\sigma_D(x, \xi)$ is invertible for every $\xi \neq 0$. Its *analytic index* is its Fredholm index on Sobolev completions; its *topological index* is the K-theoretic pushforward of its symbol class.

Exercise 7.19 (Elliptic regularity and Fredholmness)**.** Assume the construction of Sobolev completions, the Rellich compactness theorem, elliptic regularity, and, for an elliptic operator D of order m, the estimates

$$\|s\|_{H^{r+m}} \le C_r(\|Ds\|_{H^r} + \|s\|_{H^r})$$

for D and its formal adjoint. Prove that $D : H^{r+m}(E) \to H^r(F)$ is Fredholm and that its index is independent of r.

Exercise 7.20 (Atiyah–Singer index theorem)**.** Use the Thom isomorphism above, and assume the construction of the K-theoretic pushforward and uniqueness of an index transformation satisfying excision, multiplicativity, and Bott normalization. Prove that analytic index has these three properties. Verify them for topological index and conclude $\operatorname{ind}_{an} D = \operatorname{ind}_{top}(\sigma_D)$.

Definition 7.15 (Spin structures and Dirac operators)**.** A *spin structure* on an oriented Riemannian n-manifold is a lift of its oriented orthonormal frame bundle through $\mathrm{Spin}(n) \to SO(n)$. Its complex spin representation defines the *spinor bundle* S, with Clifford action c and spin connection. The *Dirac operator* is $D = c \circ \nabla^S$. In even dimensions the chirality operator splits $S = S^+ \oplus S^-$. In formal curvature roots,

$$\widehat{A}(TM) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}.$$

Exercise 7.21 (Hirzebruch–Riemann–Roch)**.** Assume that the topological index of the Dolbeault symbol on a compact complex manifold is $\int_M \mathrm{ch}(E)\,\mathrm{Td}(TM)$, where $\mathrm{Td}(TM) = \prod_j x_j/(1 - e^{-x_j})$ in formal Chern roots. Apply Atiyah–Singer to derive the Hirzebruch–Riemann–Roch formula and recover Riemann–Roch for a line bundle on a compact Riemann surface.

Exercise 7.22 (Dirac index formula)**.** Assume that the topological index of the coupled chiral Dirac symbol is $\int_M \widehat{A}(TM)\,\mathrm{ch}(E)$. Apply Atiyah–Singer to derive

$$\mathrm{ind}(D^+ \otimes E) = \int_M \widehat{A}(TM)\,\mathrm{ch}(E).$$

Definition 7.16 (Chirality and instanton number)**.** Let M be an oriented Riemannian spin four-manifold with the spinor splitting above. For a Hermitian vector bundle E with unitary connection A, the coupled Dirac operator is odd and has a *chiral part*

$$D_A^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E).$$

For an instanton or anti-instanton from Exercise 3.10, its *instanton number* is

$$k = -\frac{1}{8\pi^2}\int_M \mathrm{Tr}(F_A \wedge F_A) \in \mathbb{Z}.$$

Exercise 7.23 (Fermion chirality detects instanton number)**.** Identify the characteristic number $k(A)$ from Exercise 3.10 with $\int_M \mathrm{ch}_2(E)$. Then apply the Dirac-index formula from the preceding exercise to prove

$$\mathrm{ind}\, D_A^+ = \dim\ker D_A^+ - \dim\ker(D_A^+)^* = \int_M \big[\widehat{A}(TM)\,\mathrm{ch}(E)\big]_4.$$

For the fundamental $SU(2)$ bundle over S^4, show that this index is k with the orientation convention in the definition. Use the Weitzenböck formula for a self-dual or anti-self-dual connection to determine which chirality has no zero modes and deduce that the other has exactly $|k|$ zero modes. For a fermion in a representation R, derive the factor $2T(R)k$. Finally, compute the fermionic measure under the axial rotation $\psi \mapsto e^{i\alpha\gamma_5}\psi$ and obtain the integrated anomaly $\Delta Q_5 = 2\,\mathrm{ind}\, D_A^+$, identifying the chirality change carried by an instanton transition.

Exercise 7.24 (Index theory and invertible free-fermion phases)**.** Let $H(k)$ be a continuous family of finite-dimensional self-adjoint free-fermion Hamiltonians over a compact momentum space X, with a uniform gap at zero. Flatten its spectrum to

$$Q(k) = \mathrm{sgn}\, H(k) = H(k)|H(k)|^{-1}, \qquad P(k) = \frac{1 - Q(k)}{2},$$

and prove that the occupied spaces $E_k = \mathrm{im}\, P(k)$ form a vector bundle $E \to X$. Show that its stable class $[E] \in K^0(X)$ is invariant under gap-preserving deformations.

When X is an even-dimensional spin manifold, pair this class with its Dirac operator and use Atiyah–Singer to obtain

$$\langle [E], [D_X] \rangle = \mathrm{ind}(D_X^+ \otimes E) = \int_X \widehat{A}(TX)\,\mathrm{ch}(E).$$

For a two-dimensional Brillouin torus, identify this integer with the first Chern number and derive the class-A Hall response $\sigma_H = (e^2/h)\langle c_1(E), [X]\rangle$.

Prove that stacking Hamiltonians gives direct sum and addition in $K^0(X)$. After passing to reduced K-theory, show that spectral reversal replaces E by $E^\perp$ and gives the additive inverse because $E \oplus E^\perp$ is trivial. Explain why an invertible phase can have protected boundary modes or symmetry defects but no nontrivial intrinsic bulk fusion sectors. Contrast this group-valued classification with the sector-valued noninvertible phases in the next section.

Exercise 7.25 (The free-fermion periodic table of topological insulators and superconductors)**.** Let $H(k)$ be a spectrally flattened Bloch or Bogoliubov–de Gennes Hamiltonian on the compactified momentum space S^d. Time-reversal, particle-hole, and chiral symmetries act by

$$\mathsf{T}H(k)\mathsf{T}^{-1} = H(-k), \qquad \mathsf{C}H(k)\mathsf{C}^{-1} = -H(-k), \qquad \mathsf{S}H(k)\mathsf{S}^{-1} = -H(k),$$

where T and C are antiunitary, with square $+1$ or -1 when present, and S is unitary, with its phase chosen so that $\mathsf{S}^2 = 1$. Organize the complex classes A, AIII and the real classes in the order

s	0	1	2	3	4	5	6	7
class	AI	BDI	D	DIII	AII	CII	C	CI
T^2	$+1$	$+1$	0	-1	-1	-1	0	$+1$
C^2	0	$+1$	$+1$	$+1$	0	-1	-1	-1

and verify that when both antiunitary symmetries occur their product gives a chiral symmetry, up to an irrelevant phase.

Assume the Atiyah–Bott–Shapiro identification of stable symmetry-compatible Hamiltonians with Clifford-extension spaces, as used by Kitaev, *Periodic table for topological insulators and superconductors*. In the complex classes the strong classification groups in dimension d are

$$\pi_0(C_{-d}) \quad \text{for A}, \qquad \pi_0(C_{1-d}) \quad \text{for AIII},$$

and in real class s the group is

$$\pi_0(R_{s-d}) \cong KO^{d-s}(\mathrm{pt}).$$

Indices are periodic, and their coefficient groups are

$$\pi_0(C_q) = \begin{cases} \mathbb{Z} & q \equiv 0 \pmod 2, \\ 0 & q \equiv 1 \pmod 2, \end{cases}$$

q mod 8	0	1	2	3	4	5	6	7
$\pi_0(R_q)$	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0.

Use these data, rather than a memorized chart, to fill every entry of the periodic table:

class	T^2	C^2	S	$d=0$	1	2	3	4	5	6	7
A	0	0	0								
AIII	0	0	1								
AI	$+1$	0	0								
BDI	$+1$	$+1$	1								
D	0	$+1$	0								
DIII	-1	$+1$	1								
AII	-1	0	0								
CII	-1	-1	1								
C	0	-1	0								
CI	$+1$	-1	1								

Recover in particular the integer quantum Hall phase in class A, the one-dimensional Majorana chain in class D, the two-dimensional quantum spin Hall phase in class AII, and the three-dimensional class-DIII topological superconductor. Explain why an abstract $\mathbb{Z}$ entry coming from R_4 is often printed as $2\mathbb{Z}$ when its generator maps to twice the corresponding complex invariant; in dimension zero this is the even complex dimension forced by Kramers degeneracy. Finally, explain why this is only the table of *strong* phases: a lattice Brillouin torus can also carry weak invariants. Relate the symmetry-free case to the earlier decomposition of $K^*(\mathbb{T}^d)$, and state why antiunitary symmetries require real, equivariant, or twisted K-theory, following Freed–Moore, *Annales Henri Poincaré* **14** (2013). State why this stable noninteracting classification need not remain unchanged in the presence of strong interactions.

Part VII

Anyons

8 Anyons

Experimental observations.

- Modulation spectroscopy of critical Rydberg chains resolves even-parity excitation ratios $2:4:6:8$ and odd-parity ratios $3:5:7$ associated with the Ising conformal spectrum.
- At filling $\nu = 1/3$, shot-noise measurements give quasiparticle charge near $e/3$, and interferometry detects phase evolution associated with fractional statistics.

8.1 Subfactors and Conformal Nets

Subfactor index measures the size of an inclusion of observable algebras. For conformal nets, finite index organizes localized charges into a finite system of superselection sectors.

Definition 8.1 (Subfactors, conditional expectations, and index)**.** A *subfactor* is an inclusion $N \subseteq M$ of factors with common unit. A *normal conditional expectation* $E : M \to N$ is a normal positive unital N-bimodule projection. For finite factors with normalized trace, the *Jones index* $[M : N]$ is the Murray–von Neumann dimension of $L^2(M)$ as a left N-module, and $E_N : M \to N$ denotes the unique trace-preserving normal conditional expectation.

For arbitrary factors and a faithful normal conditional expectation E, set

$$[M : N]_E = \lambda_E^{-1}, \qquad \lambda_E = \sup\big(\{0\} \cup \{\lambda > 0 : E(x) \geq \lambda x \text{ for every } x \in M_+\}\big).$$

The *minimal index* is $[M : N]_{\min} = \inf_E [M : N]_E$, with value $+\infty$ if no finite-index expectation exists. For the type-III inclusions below, $[M : N]$ denotes this minimal index.

Definition 8.2 (Basic construction)**.** For finite factors $N \subseteq M$, the *Jones projection* e_N on $L^2(M)$ is the orthogonal projection onto $L^2(N)$. The *basic construction* is $M_1 = \langle M, e_N\rangle''$. Iteration gives the *Jones tower* $N \subseteq M \subseteq M_1 \subseteq M_2 \subseteq \cdots$.

Exercise 8.1 (Jones tower and Temperley–Lieb relations)**.** For a finite-index inclusion $N \subseteq M$ of finite factors, derive $e_N x e_N = E_N(x) e_N$ and construct the canonical trace on the basic construction. Prove that the normalized Jones projections in the tower satisfy

$$e_i^2 = e_i, \qquad e_i e_j = e_j e_i \ (|i - j| > 1), \qquad e_i e_{i\pm 1} e_i = [M : N]^{-1} e_i.$$

Exercise 8.2 (Diagrammatic calculus and index quantization)**.** Construct the Temperley–Lieb diagrammatic category, with composition by stacking and each closed loop valued at $\delta = [M : N]^{1/2}$. Use positivity of its Markov trace to prove that index values below 4 are $4\cos^2(\pi/n)$ for integers $n \geq 3$.

Definition 8.3 (Braid groups and Markov traces)**.** The *braid group* B_n has generators σ_i with relations $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ and $\sigma_i\sigma_j = \sigma_j\sigma_i$ for $|i-j| > 1$. A *Markov trace* on a tower of braid-group quotients is a compatible trace satisfying a fixed stabilization rule.

Exercise 8.3 (Jones polynomial)**.** Construct a braid-group representation from the Temperley–Lieb generators. Use the Markov trace and Markov moves to obtain an invariant of oriented links, normalize it to the Jones polynomial, and derive its skein relation. Compute it for the unknot, trefoil, and Hopf link.

Definition 8.4 (Local conformal nets)**.** Let $\mathcal{I}$ be the proper open intervals of S^1. A *local conformal net* assigns a von Neumann algebra $\mathcal{A}(I) \subset B(H)$ to each $I \in \mathcal{I}$ such that

$$I \subset J \Longrightarrow \mathcal{A}(I) \subset \mathcal{A}(J), \qquad I \cap J = \varnothing \Longrightarrow [\mathcal{A}(I), \mathcal{A}(J)] = 0.$$

The assignment is covariant under orientation-preserving diffeomorphisms of S^1, represented projectively on H, and the rotation generator is positive. Its vacuum vector Ω is invariant under the Möbius subgroup and cyclic for the local algebras, and $\bigvee_I \mathcal{A}(I) = B(H)$. Physically, S^1 is the compactified chiral light ray and $\mathcal{A}(I)$ consists of observables localized in I. The norm-closed C^*-algebra

$$\mathfrak{A} = C^*\left(\bigcup_{I\in\mathcal{I}} \mathcal{A}(I)\right)$$

generated by the local algebras is the *quasilocal algebra*. For the regular nontrivial conformal nets used below, the local algebras are hyperfinite type-III_1 factors; the physical content lies in their inclusions and covariance, not in their abstract isomorphism type alone.

Definition 8.5 (Duality and regularity for conformal nets)**.** Write I' for the interior of the complement of I. A net has *Haag duality* when $\mathcal{A}(I) = \mathcal{A}(I')'$. It has the *split property* when $\overline{I} \subset J$ implies $\mathcal{A}(I) \subset F \subset \mathcal{A}(J)$ for some type-I factor F. It is *strongly additive* when removing an interior point from I to obtain $I_1 \cup I_2$ gives $\mathcal{A}(I) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. A representation of the quasilocal algebra is *locally normal* when its restriction to every $\mathcal{A}(I)$ is normal.

Exercise 8.4 (The two-interval subfactor)**.** For $E = I_1 \cup I_2$ a union of disjoint intervals, set $\mathcal{A}(E) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. First use locality and vacuum cyclicity for $\mathcal{A}(I')$ to prove that the vacuum is separating for $\mathcal{A}(I)$. Use the split property and the spatial tensor product to identify separated local algebras as independent subsystems. Then use locality to obtain the inclusion

$$\mathcal{A}(E) \subseteq \mathcal{A}(E')'.$$

Under the split and irreducibility hypotheses, show that this is a subfactor. Define the *μ-index* $\mu(\mathcal{A}) = [\mathcal{A}(E')' : \mathcal{A}(E)]$, and prove that it is independent of the two-interval configuration by conformal covariance. Contrast single-interval Haag duality $\mathcal{A}(I) = \mathcal{A}(I')'$ with the possible failure of duality for E, measured by this Jones index.

Definition 8.6 (DHR sectors)**.** A *DHR sector* of $\mathcal{A}$ is represented by a locally normal, transportable endomorphism ρ of the quasilocal algebra that acts identically on $\mathcal{A}(I')$ for some interval I; transportability means that an equivalent representative can be localized in any interval. Composition is the tensor product. For a finite-index irreducible sector localized in I, its statistical dimension satisfies

$$d(\rho)^2 = [\mathcal{A}(I) : \rho(\mathcal{A}(I))].$$

Exercise 8.5 (Critical Ising chain, Rydberg spectroscopy, and the Ising net)**.** For $J, h > 0$, consider the transverse-field Ising chain

$$H_{\mathrm{I}} = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i.$$

Use the Jordan–Wigner transformation and a Bogoliubov rotation to obtain

$$\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh\cos k}.$$

Show that the gap is $2|J-h|$ and closes at $h = J$, where the long-wavelength modes are massless Majorana fermions. Taking the standard fermionic determinant asymptotics as input, obtain $\langle Z_i Z_{i+r}\rangle \sim Cr^{-1/4}$ and a connected energy-density correlator proportional to r^{-2}. Identify the bulk Ising fields $1, \sigma, \varepsilon$ with scaling dimensions $0, 1/8, 1$, respectively.

The Rydberg experiment realizes a different microscopic chain,

$$H_{\mathrm{R}} = \sum_i \left(\frac{\Omega}{2} P_{i-1} X_i P_{i+1} - \Delta n_i \right) + \sum_{j-i \geq 2} V_{ij} n_i n_j,$$

in the nearest-neighbor blockade regime. Explain why equality of microscopic Hamiltonians is unnecessary for the two critical points to share the Ising universality class; here n_i projects onto the Rydberg state and $P_i = 1 - n_i$. Take as input the open-chain CFT spectrum, including its conformal towers and $E_a(L) - E_0(L) = \pi v x_a / L + o(L^{-1})$, and recover the universal even-parity ratios $2:4:6:8$ and odd-parity ratios $3:5:7$. Compare them with the reported modulation-spectroscopy peaks of Sun et al., *Experimental observation of conformal field theory spectra* (2026), stating the finite-size and experimental limitations.

Identify the chiral continuum theory with the $c = 1/2$ Ising conformal net. Its DHR sectors $1, \psi, \sigma$ have conformal weights $0, 1/2, 1/16$; relate the bulk energy field to $\psi\bar{\psi}$ and distinguish these chiral weights from the bulk scaling dimensions above. Explain why the experiment supports the Ising CFT spectrum but does not directly measure the net or all of its superselection sectors.

8.2 Fusion Categories and Topological Field Theory

The sector operations abstract to a braided fusion category. Its diagrammatic data in turn define topological field theories and link invariants.

Definition 8.7 (Monoidal and rigid categories)**.** A *monoidal category* is a category $\mathcal{C}$ with a tensor functor, unit object, and coherent associativity and unit isomorphisms. A *dual* of X is an object X^* with evaluation and coevaluation morphisms satisfying the snake identities. A monoidal category is *rigid* if every object has left and right duals.

Definition 8.8 (Fusion, braided, and modular categories)**.** A *fusion category* over $\mathbb{C}$ is a finite semisimple rigid $\mathbb{C}$-linear monoidal category with simple unit. It is *unitary* when it is a C^*-category and its tensor product and structural isomorphisms are compatible with the adjoint. A *braiding* is a natural family $c_{X,Y} : X \otimes Y \to Y \otimes X$ satisfying the hexagon identities. A *unitary modular tensor category* is a unitary ribbon fusion category with unitary nondegenerate braiding.

Exercise 8.6 (From Jones index to a modular category)**.** Assume that $\mathcal{A}$ is completely rational: it is split and strongly additive, with finite μ-index. Assume that charge transporters for localized finite-index DHR endomorphisms produce a unitary braiding and that the complete-rationality theorem makes their category a unitary modular tensor category with finitely many irreducible sectors ρ_i and

$$\mu(\mathcal{A}) = \sum_i d(\rho_i)^2.$$

Identify composition with tensor product, conjugate endomorphisms with duals, and charge transporters with the braiding. Explain where nondegeneracy enters modularity. Use Kawahigashi–Longo–Müger, *Communications in Mathematical Physics* **219** (2001). For the Ising net at $c = 1/2$, take the sectors $1, \psi, \sigma$ with dimensions $1, 1, \sqrt{2}$ and fusion rules

$$\psi^2 = 1, \qquad \psi\sigma = \sigma, \qquad \sigma^2 = 1 \oplus \psi.$$

Compute $\mu(\mathcal{A})$ and interpret ψ and σ as the chiral Majorana and spin sectors.

Exercise 8.7 (Fusion rules and braiding)**.** Prove that simple-object classes of a fusion category form a based ring under tensor product. Derive the pentagon and hexagon equations for associativity and braiding data. Compute quantum dimensions and braid-group representations for the Fibonacci fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$.

Definition 8.9 (Bordisms and topological field theories)**.** The *bordism category* has closed oriented $(n-1)$-manifolds as objects, oriented n-dimensional bordisms as morphisms, and disjoint union as tensor product. An *n-dimensional topological field theory* is a symmetric monoidal functor from this category to vector spaces. An extended theory may assign categories to codimension-two manifolds. A TQFT is *invertible* when its state spaces are tensor-invertible (hence one-dimensional over a field) and every bordism acts invertibly; noninvertible TQFTs may instead carry nontrivial superselection and fusion sectors.

Definition 8.10 (The Reshetikhin–Turaev $SU(2)_k$ theory)**.** For $k \in \mathbb{N}$, let $\mathcal{C}_k = SU(2)_k$ be the unitary modular tensor category with simple objects $V_0, \ldots, V_k$, where V_j has spin $j/2$. The Reshetikhin–Turaev construction uses the colored ribbon-graph calculus of $\mathcal{C}_k$ and surgery to define a symmetric monoidal $(2+1)$-dimensional TQFT

$$Z_k^{\mathrm{RT}} = Z_{\mathcal{C}_k}$$

on bordisms equipped with the extra framing data needed to cancel the framing anomaly. It assigns invariants to closed framed three-manifolds and to framed links colored by the objects V_j. We call Z_k^{RT} the Reshetikhin–Turaev model of quantum $SU(2)$ Chern–Simons theory at level k; this terminology does not define it by a functional integral.

Definition 8.11 (Chern–Simons theory and Wilson loops)**.** Let G have an invariant inner product on $\mathfrak{g}$, and fix a trivialization of a principal G-bundle over a closed oriented three-manifold. For the resulting $\mathfrak{g}$-valued connection potential A, the *Chern–Simons functional* at level k is

$$S_{CS}(A) = \frac{k}{4\pi} \int_M \left(\langle A \wedge dA \rangle + \frac{1}{3} \langle A \wedge [A \wedge A] \rangle \right).$$

For a representation V and closed curve K, the *Wilson loop* is $W_V(K) = \mathrm{Tr}_V(\mathrm{Hol}_A(K))$.

Exercise 8.8 (Gauge invariance, level, and knot invariants)**.** For compact, connected, simply connected simple G, normalize the inner product so that the winding term of a gauge transformation lies in $2\pi\mathbb{Z}$. Compute the change of S_{CS} and derive $k \in \mathbb{Z}$ as the condition for $e^{iS_{CS}}$ to be gauge invariant. Prove that its critical points are flat connections.

Exercise 8.9 (Reshetikhin–Turaev links and Jones evaluations)**.** Use the Temperley–Lieb realization of $\mathcal{C}_k$ and its ribbon functor to show that the Z_k^{RT} invariant of a framed link in S^3, with a component colored by V_j, is the corresponding j-colored Jones evaluation at

$$q = e^{2\pi i/(k+2)}.$$

In particular, identify V_1 with the fundamental color giving the ordinary Jones polynomial. Compute the normalized invariants of the unknot, Hopf link, and trefoil, state the normalization of the colored unknot, and track the framing factor separately from the ambient-isotopy invariant.

Exercise 8.10 (Comparison with Chern–Simons quantization)**.** Do not use the formal Chern–Simons path integral as a definition. Assume the comparison theorem identifying the modular functor underlying Z_k^{RT} with the level-k $SU(2)$ Wess–Zumino–Witten conformal-block modular functor, the quantization associated with $SU(2)$ Chern–Simons theory; see Andersen–Ueno, *Construction of the Witten–Reshetikhin–Turaev TQFT from conformal field theory*. Explain why this gives a precise sense in which Z_k^{RT} realizes quantum Chern–Simons theory and matches its Wilson-loop labels and gluing rules. Also explain why the comparison theorem does not by itself construct a nonperturbative measure for the formal path integral

$$\int \mathcal{D}A\, e^{iS_{CS}(A)} \prod_r W_{V_{j_r}}(K_r).$$

Exercise 8.11 (Functorial gluing)**.** Show that the monoidal-functor axioms give multiplicativity under disjoint union, composition under gluing, duality under orientation reversal, and mapping-class-group representations on state spaces. In two dimensions, prove the equivalence with commutative Frobenius algebras.

8.3 Topological Order and Quantum Codes

In a gapped two-dimensional phase, the same categorical data describe anyon fusion, braiding, topology-dependent state spaces, and protected logical observables.

Definition 8.12 (Anyons and topological degeneracy)**.** In a two-dimensional gapped phase, an *anyon type* is a simple object of its braided fusion category. It is *abelian* if its quantum dimension is 1 and *nonabelian* otherwise. The *topological degeneracy* on a closed surface is the dimension of the state space assigned by the associated topological field theory.

Exercise 8.12 (The toric code: error correction from topological order)**.** Give an $L \times L$ square cellulation of the torus, with $L \geq 2$, a qubit on every edge. Let X and Z be self-adjoint unitaries satisfying $X^2 = Z^2 = 1$ and $XZ = -ZX$ on one qubit. Operators on distinct edges commute. For each vertex v and plaquette p, set

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \subset \partial p} Z_e, \qquad H = -\sum_v A_v - \sum_p B_p.$$

Prove that all A_v and B_p commute and that the ground space is their common $+1$ spectral subspace. Establish the two relations $\prod_v A_v = 1$ and $\prod_p B_p = 1$, prove that the remaining stabilizers are independent, find $E_0 = -2L^2$, and compute

$$\dim \ker(H - E_0) = 2^{2L^2 - (L^2 + L^2 - 2)} = 4.$$

Show that an open Z-string creates violated vertex terms only at its endpoints, while an open dual X-string creates violated plaquette terms only at its endpoints. Prove that contractible closed strings are products of stabilizers. Show that strings around the two noncontractible cycles give two pairs of logical operators, with a primal Z-loop and a dual X-loop anticommuting exactly when their intersection number is odd. Deduce that the code encodes two logical qubits and has distance L.

Let $\mathcal{S}$ be the stabilizer group generated by the A_v and B_p, let $\mathcal{S}' \subseteq B(H)$ be its commutant, and let P be the ground space projection. Prove that the logical observable algebra is

$$\mathcal{A}_{\log} = P\mathcal{S}'P = \operatorname{span}\{P\overline{X}_1^{i_1}\overline{Z}_1^{j_1}\overline{X}_2^{i_2}\overline{Z}_2^{j_2}P : i_r, j_r \in \{0,1\}\} \cong M_2(\mathbb{C}) \otimes M_2(\mathbb{C}),$$

where the barred operators are noncontractible logical loops. Equivalently, identify the logical Pauli group as the Pauli centralizer of $\mathcal{S}$ modulo $\mathcal{S}$. If an error channel has Kraus operators E_a for which every $E_a^*E_b$ is supported on fewer than L edges, prove that

$$PE_a^*E_bP \in \mathbb{C}P = \mathcal{A}_{\log}'$$

and hence verify the exact OAQEC condition for $\mathcal{A}_{\log}$. Deduce in particular that the toric code corrects arbitrary errors on at most $\lfloor (L-1)/2 \rfloor$ edges.

Interpret the endpoint excitations as anyons e and m. Derive $e^2 = m^2 = 1$, set $\varepsilon = em$, and show that taking e once around m produces the phase -1. Explain how local error syndromes, global logical Wilson loops, and the fourfold TQFT degeneracy express the same topological structure; see Kitaev, *Fault-tolerant quantum computation by anyons*, and Dauphinais–Kribs–Vasmer, *Stabilizer Formalism for Operator Algebra Quantum Error Correction*, Quantum **8** (2024).

Exercise 8.13 (Abelian and nonabelian anyons)**.** Compute fusion spaces and braid actions for an abelian category and for the Fibonacci category. Prove that nonabelian fusion spaces can have dimension greater than one and that braiding gives linear braid-group representations on them. Explain why the corresponding physical gates are commonly regarded projectively after global phases are discarded. Compute the torus degeneracy from the simple anyon types in a modular category.

Exercise 8.14 (Quantum Hall phases and measured topological order)**.** Use flux insertion for a Laughlin state at filling $\nu = 1/m$ to derive quasiparticle charge e/m, exchange phase π/m, Hall conductance $\nu e^2/h$, and genus-g degeneracy m^g. Compare these consequences separately with conductance plateaus, fractional shot-noise charge measurements, and interferometric phase data, stating which topological predictions each observation tests and which it does not.